Project Assessment: Foundations of AI – FoodHub Low Code Project

MIT Applied AI and Data Science Program

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December 16, 2025

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Executive Summary

The FoodHub food aggregator company, located in New York City, operates in a densely populated area that offers key advantages, including a wide range of Restaurants, high customer demand, and substantial Weekend Order volume. The FoodHub operates using a smartphone app that allows customers to place direct orders to a wide range of restaurants, while also using designated delivery people to pick up the Order from a restaurant and drop it off at the customer's location. Customers can then leave a Rating based on the service they received. The FoodHub company then earns a fixed-margin revenue from delivery orders placed by restaurants. Thus, data analysis of Order Costs, customer Rating behavior, and business-processed data reveals opportunities to boost customer satisfaction, streamline operations, and amplify Revenue through focused marketing, advanced logistics, and strengthened Restaurant partnerships.

Conclusions

The company sees strong Weekend demand, driven by a strong inclination toward American cuisine among its five top-performing Restaurants. The company should focus on developing strong partnerships with the top-performing Restaurants in its marketplace (e.g., Shake Shack and The Meatball Shop). Delivery Times are efficient, with orders averaging less than 30 minutes. Many Ratings are missing, causing an inaccurate customer satisfaction analysis. There is a strong baseline of positive experiences, driven by an abundance of 5-star Ratings. The current Total Net Revenue from the dataset is about $6,166.3, where most orders do not surpass $20. We will later see observations, visualizations, and implications that justify these conclusions.

Actionable Insights

According to the dataset, there are 20 top-performing Restaurants. These Restaurants are responsible for high order count and Revenue. Delivery Times on Weekends (22 minutes) are faster than on Weekdays (28 minutes). As a whole, the majority of orders (89.46%) do not take longer than 60 minutes to deliver. Most customers order food in Restaurants that offer American, Japanese, and Italian cuisines. Customer Ratings drop whenever delivery times exceed about 25 minutes,

suggesting that delivery times strongly correlate with customer satisfaction. Restaurants (e.g., Meatball Shop, Blue Ribbon Fried Chicken, and Shake Shack) with high Ratings are preferred for promotional campaigns. A handful of Restaurants are responsible for the total net Revenue. The total amount of missing Ratings affects the credibility of customer satisfaction parameters. We will later see observations, visualizations, and implications that justify these actionable insights.

Recommendations

FoodHub should launch a loyalty program and introduce upsell bundling on its products to attract more customers. FoodHub should also consider using marketing resources and campaigns to promote top-performing Restaurants, increasing customer retention and acquisition, brand awareness, and Revenue. As part of marketing and campaigns, FoodHub should promote high-demand dishes by highlighting them. Additionally, these campaigns should emphasize Weekends, given the higher demand than on Weekdays. Chiefly, FoodHub should also create premium offers and subscription plans for its customers, and partner with high-performing Restaurants to optimize Revenue. Given the high number of "Not given" Ratings, FoodHub should implement in-app prompts and push notifications in its smartphone app and follow-up surveys to reduce "Not given" Ratings and increase participation in Ratings. Also, FoodHub should consider expanding kitchen staffing across Restaurants during the week to minimize delays in food preparation. Within the same scope, FoodHub should implement an optimized navigation algorithm to potentially lower delivery times, especially for deliveries that take an hour or longer.

Key Conclusions and Recommendations to Improve FoodHub Business

The FoodHub company's business shows the following analytical conclusions, driven by Cuisine Types and Rating data characteristics and findings. High customer demand for American cuisine occurs on Weekends, with over 400 orders, indicating that most customers prefer it. Most customers also order on Weekends rather than Weekdays, suggesting they prefer leisure dining and social gatherings. As mentioned before, a large number of Ratings are missing, with Ratings of “5” dominating over Ratings of “3” and “4”. Altogether, this suggests that most customers are satisfied with their service. Still, some customers might be dissatisfied due to longer total Delivery Times, caused by increased Food Preparation Time and Delivery Time delays. Also,

top-rated Restaurants such as The Meatball Shop, Blue Ribbon Fried Chicken, and Shake Shack have strong reputations, with customers showing strong loyalty. Most deliveries take an average of 28 minutes on Weekdays and 22 minutes on Weekends. Approximately 1 in 10 orders takes longer than 60 minutes to deliver, whereas Food Preparation Time across Cuisine Types ranges from 24 to 31 minutes. These operational insights suggest Weekday gridlocks, a significant improvement zone for customer experience based on delivery time delays, and the need to establish a baseline for Food Preparation Times. As for Revenue, top-rated and performing Restaurants, such as Shake Shack and The Meatball Shop, are seen as sales generators. In contrast, mid-tier Restaurants show potential for upselling and advertising support. The top frequent high-value customers offer suggestions for incorporating loyalty programs and can serve as predictive indicators of demand.

The FoodHub company's business shows the following analytical recommendations, driven by Cuisine Types and Rating data characteristics and findings. FoodHub should focus on promoting American cuisine options on Weekends to increase Revenue. Restaurants that offer higher-cost Cuisine Types, such as Japanese and Italian, should be featured in premium advertising, and Restaurants that offer lower-cost Cuisine Types, such as Vietnamese and Korean, should be featured as affordable, fast-food-prep alternatives. FoodHub should also provide incentives, such as discounts, to encourage customers to leave Ratings. As mentioned before, in-app notifications an prompts could alert more customers to leave Ratings. FoodHub should also account for the fact that each Restaurant has sufficient Restaurant employees, and delivery drivers on Weekdays to lower Weekday Delivery Times. FoodHub should also implement strong restaurant partnership programs with top-rated Restaurants, such as Shake Shack and The Meatball Shop, leveraging loyalty programs and premium product offerings. FoodHub could also incorporate bundled meals and combo pricing at lower- to mid-tier Restaurants as effective partnership options for Restaurants. FoodHub should also offer top-frequent-customer outreach programs, including exclusive offers and early access to new programs available only to customers with loyalty subscriptions. FoodHub should also use strategic pricing across Restaurants, featuring combo deals in the $12- $15 range, and incorporate premium food pricing options aligned with high-paying customer expectations.

Business Problem Overview and Solution Approach

Business Problem Overview

FoodHub connects customers to a multitude of Restaurants across New York City via a smartphone app. Even though there is a growing demand for online delivery among its customers (e.g., students and busy professionals), the company faces the following challenges that inhibit growth and customer satisfaction. These challenges include customer retention and engagement, where many customers fail to return to using the app and do not leave a Rating. There are inconsistencies in Delivery Times, with about 1 in 10 (10%) orders exceeding 60 minutes. The commission fees within each Order are high, which reveals disapproval among about 30% of partnering Restaurants. There is a need for Revenue optimization, as fewer than one-third of all orders exceed $20. Also, a few top-performing Restaurants account for most of the Revenue, whereas most Restaurants fall short.

Solution Approach

FoodHub should follow a data-driven, multi-faceted methodology, using Python-driven statistical modeling and computations conducted by the Data Science team to address the above critical problems and to reveal key insights, operational findings, strategies, and recommendations from the provided dataset. The approach emphasizes Python-powered statistical analyses (e.g., code snippets are found in the appendix) to deliver comprehensive data analysis and insights to support strategic interventions and recommendations. Relating to data analysis and insights, the data science team investigated customer clustering behavioral metrics (e.g., identifying top customers and cuisine preferences), Restaurant performance benchmarking (e.g., comparing total Number of Orders, Revenue, and Ratings across Restaurants), and operational metrics (e.g., tracking Food Preparation and Delivery Times). Relating to data analysis and insights, the data science team investigated customer clustering behavioral metrics (e.g., identifying top customers and cuisine preferences), restaurant performance benchmarking (e.g., comparing total Number of Orders, Revenue, and Ratings across Restaurants), and operational metrics (e.g., tracking Food Preparation and Delivery Times). Lastly, the overall methodological approach focuses on delivering projections that address inferences on customer retention, delivery efficiency, Restaurant partnering satisfaction, and revenue optimization.

Data Overview

In this section, the structure of the data from the provided dataset is presented. The dataset collected by FoodHub includes 1,898 unique Orders (each represented by a row).

Categorical Variables

The dataset collected by FoodHub includes 1,898 unique Orders consists of the following categorical variables: Order ID (e.g., the unique identifier for each Order), Customer ID (e.g., the ID of the customer placing an order), Restaurant Name (e.g., the name of the Restaurant fulfilling the Order), Cuisine Type (e.g., the name of the kind of cuisine being ordered), and Day of the Week (e.g., identifies if an order was placed on a Weekday–Monday, Tuesday, Wednesday, Thursday, Friday, or Weekend –Saturday, Sunday).

Numerical Variables

The dataset collected by FoodHub includes 1,898 unique orders consists of the following numerical variables: Cost of each Order (e.g., the Cost of each Order), Rating (e.g., indicates the customers' Rating of the service they received after receiving an Order), Food Preparation Time (e.g., the time taken for the Restaurant to prepare the food), and Delivery Time (e.g., the time taken for the delivery employee to deliver the food).

Understanding the Structure of the Data from the Dataset (See Appendix for Code Snippet)

Table 1. Sample preview of the First 5 Rows in the dataset

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24

Observations on the First 5 Rows in the Dataset

As noted earlier in the description of both categorical and numerical variables, the table shows each variable featured in separate columns from left to right as: Order

ID, Customer ID, Restaurant Name, Cuisine Type, Cost of the Order, Day of the Week, Rating, Food Preparation Time, and Delivery Time.

Shape of the Dataset (See Appendix for Code Snippet)

Code Output 1. Shape of the Dataset

(1898, 9)

Observations on the Shape of the Dataset

The dataset has 1898 rows and 9 columns. The FoodHub dataset provides a well-abundant, standardized set of total observations (e.g., the number of rows) for each column (e.g., the variables represented in each column).

Data Types found in the Dataset (See Appendix for Code Snippet)

Table 2. Column Variables with associated Data Types

```
Data columns (total 9 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   order_id               1898 non-null   int64
 1   customer_id            1898 non-null   int64
 2   restaurant_name        1898 non-null   object
 3   cuisine_type           1898 non-null   object
 4   cost_of_the_order      1898 non-null   float64
 5   day_of_the_week        1898 non-null   object
 6   rating                 1898 non-null   object
 7   food_preparation_time  1898 non-null   int64
 8   delivery_time          1898 non-null   int64
dtypes: float64(1), int64(4), object(4)
```

Observations on Dataset Types

The dataset provided initially has five numerical columns (integer and float types) and four object-type columns. Variables with integer types are represented with whole numbers, variables with float types are represented as decimals, and object types are represented as non-numeric categories. The data types for the Order ID, Customer ID, Food Preparation Time, and Delivery Time columns are integers. The

Restaurant Name, Cuisine Type, Day of the week, and Rating columns are object type variables. Lastly, the Cost of each Order column is a float type variable. Order ID and Customer ID are read as integer columns, but later converted to object columns because they are classified as identifiers. Food Preparation Time and Delivery Time are read as integer columns but can be converted to float columns for continuous calculations. The Rating column is read as an object type but later converted to an integer to ensure proper numerical calculations.

Missing Values in the Dataset (See Appendix for Code Snippet)

Table 3. Number of Missing Values in the Dataset

	0
order_id	0
customer_id	0
restaurant_name	0
cuisine_type	0
cost_of_the_order	0
day_of_the_week	0
rating	0
food_preparation_time	0
delivery_time	0

Observations on Missing Values in the Dataset

After running the appropriate null sum function, no missing values were found across all columns in the dataset. This finding suggests that FoodHub successfully compiled a complete dataset. In other words, the analysis in this report is cleaner and includes insights and conclusions with higher reliability, resulting in accurate and justified recommendations.

Summary Descriptive Statistics for Numerical Variables (See Appendix for Code Snippet)

Table 4. Summary Descriptive Statistics for Numerical Variables

	count	mean	std	min	25%	50%	75%	max
cost_of_the_order	1898.0	16.50	7.48	4.47	12.08	14.14	22.30	35.41
food_preparation_time	1898.0	27.37	4.63	20.00	23.00	27.00	31.00	35.00
delivery_time	1898.0	24.16	4.97	15.00	20.00	25.00	28.00	33.00

Descriptive Statistics for the Cost of each Order

The minimum Cost of each Order is about $4. The average Cost of each Order is about $17. The maximum Cost of each Order is about $35. The average Cost suggests a substantial mid-range point that most customers can afford, while also being sufficient for most Restaurants' profits.

Descriptive Statistics for Food Preparation Time

The minimum Food Preparation Time is 20 minutes. The average Food Preparation Time is about 27 minutes. The maximum Food Preparation Time is 35 minutes. The average Food Preparation Time suggests that Restaurants maintain a suitable pace.

Descriptive Statistics for Delivery Time

The minimum Delivery Time is 15 minutes. The average Delivery Time is 24 minutes. The maximum Delivery Time is 33 minutes. The average Delivery Time suggests efficient service, with most customers not waiting more than 30 minutes to receive their orders.

Rating unique values found in the Rating column (See Appendix for Code Snippet)

Table 5. Number of times a Rating is given across all Unique Values

	count
rating	
Not given	736
5	588
4	386
3	188

Observations on counts found across all Rating Values

There are a total of 736 orders without Ratings. There are a total of 588 orders with a given Rating of 5. There are a total of 386 orders with a given rating of 4. There are a total of 188 orders with a given Rating of 3. As mentioned before and later in the report, there are specific efforts to reduce the number of "Not given" Ratings and "3" Ratings, while increasing the number of "5" and "4" Ratings.

Univariate Analysis

A univariate analysis was conducted to examine the nature, distribution, and implications of each variable in the dataset. The median is the middle value in a dataset, splitting the data into two equal halves. The terms quartile 1 and quartile 3 indicate that the data are located in the lower 25% and upper 75% of the data shown in the following boxplots, where both lower and upper bounds represent the majority of values for distinct numerical variables.

Order ID (See Appendix for Code Snippet)

Code Output. 2 Number of Unique Order IDs

1898

Observations on Order ID

The Order ID is a unique identifier for each Order, with 1,898 unique IDs. Each Order is unique. The number of Order IDs matches the total number of orders in the dataset.

Customer ID (See Appendix for Code Snippet)

Code Output. 3 Number of Unique Customer IDs

1200

Observations on Customer ID

Each customer has a unique Customer ID. There are 1200 registered customers in the dataset, each with a unique Customer ID.

Restaurant Name (See Appendix for Code Snippet)

Code Output. 4 Number of Restaurant Names

178

Observations on the Restaurant Name

For the Restaurant Name categorical variable, there are 178 unique Restaurants, providing a variety of distinct Cuisine Types in the dataset. Later, we will see that both top-performing and underperforming Restaurants are responsible for higher and lower revenue costs, respectively.

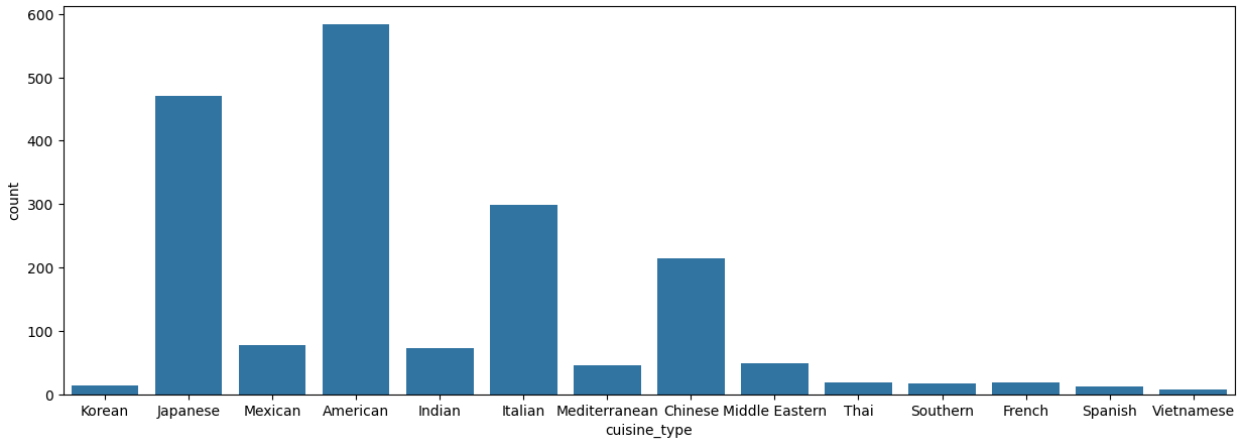
Cuisine Type (See Appendix for Code Snippets)

Code Output. 5 Number of Cuisine Types

14

For the Cuisine Type variable, there are 14 distinct cuisines in the dataset.

Graph 1. Different Cuisine Types



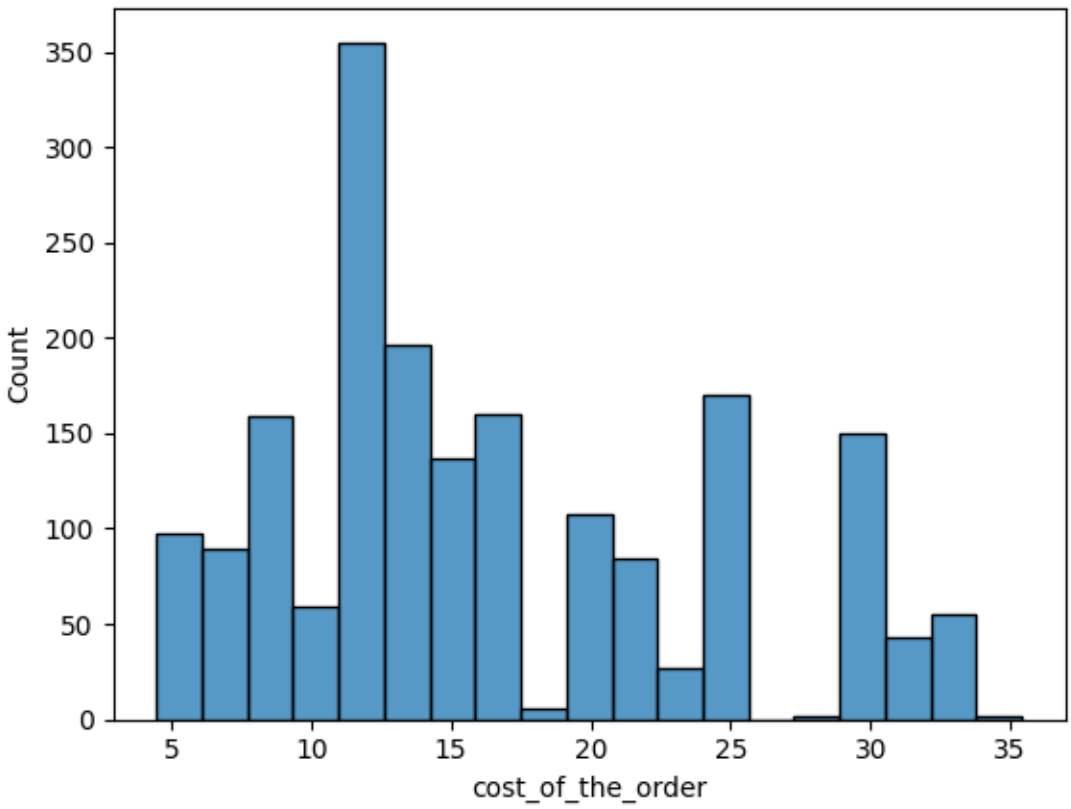
Observations on Cuisine Type Bar Graph

According to the bar graph, the most common Cuisine Type in the dataset is American, with approximately 580 orders. Japanese and Italian are the second- and third-most-common cuisines, with approximately 470 and 300 orders, respectively. Afterward, cuisines such as Chinese, Mexican, Indian, Mediterranean, and Middle Eastern are moderately represented, with approximately 220, 80, 80, 50, and 50

orders, respectively. The least represented cuisines, including Thai, Southern, French, and Spanish cuisines, each have approximately 30 orders, whereas Vietnamese and Korean have approximately 20 orders. These could be considered niche-type Restaurants. Overall, the distribution in the graph is highly skewed, as a few cuisines dominate while many others receive few orders. Based on the total number of cuisines, there is evidence of substantial variety and global appeal for all customers.

Cost of the Order (See Appendix for Code Snippets)

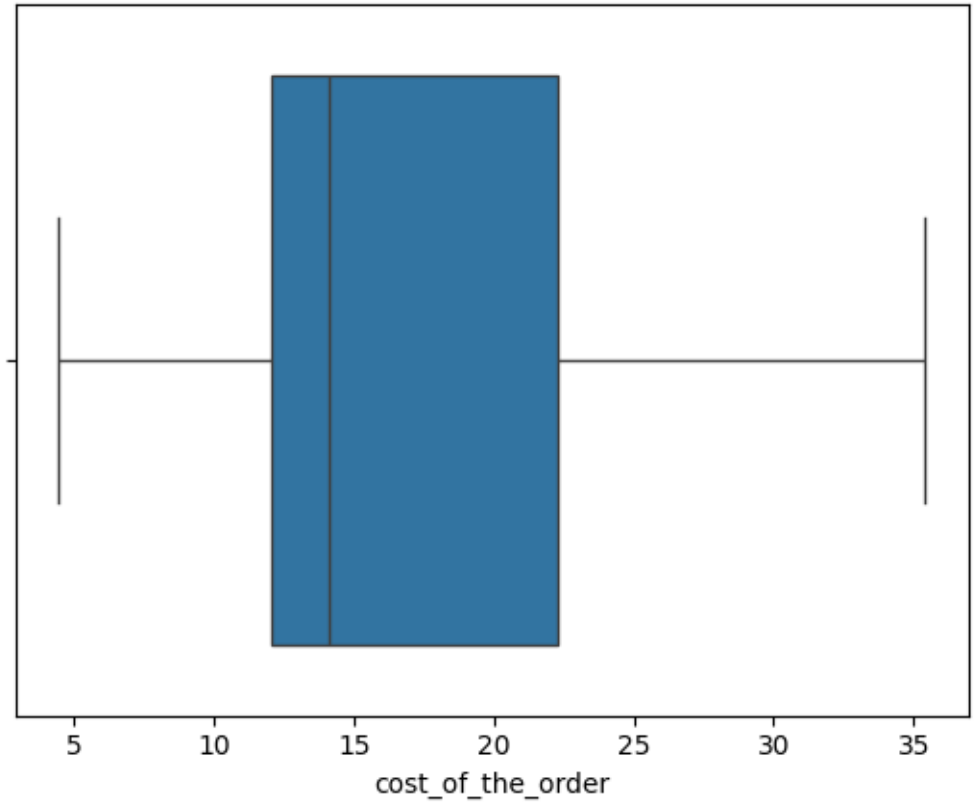
Graph 2. Cost of the Order Histogram



Observations on the Cost of the Order Histogram

The histogram shows that the distribution of Order Costs is slightly right-skewed (positively skewed). The histogram shows that the peak density is concentrated around the $10-$15 range, indicating most orders cost $10 to $15. The histogram shows secondary peaks at $25 and $30, but they are smaller, indicating secondary standard-order amounts.

Graph 3. Cost of the Order Boxplot



Observations on the Cost of the Order Boxplot

The boxplot also shows that the distribution of the Cost of each Order appears slightly right-skewed (positively skewed). The median (e.g., close to $15) is closer to the lower quartile 1 bound, which is about $12. Conversely, the boxplot shows that the upper quartile 3 bound is about $22. Given the lower quartile (1) and upper quartile (3) bounds, most order costs fall between $12 and $22. The boxplot indicates a minimum of about $5, and a maximum of about $35. There should be opportunities to upsell, bundle, and run premium promotions to increase the average order value.

Day of the Week (See Appendix for Code Snippets)

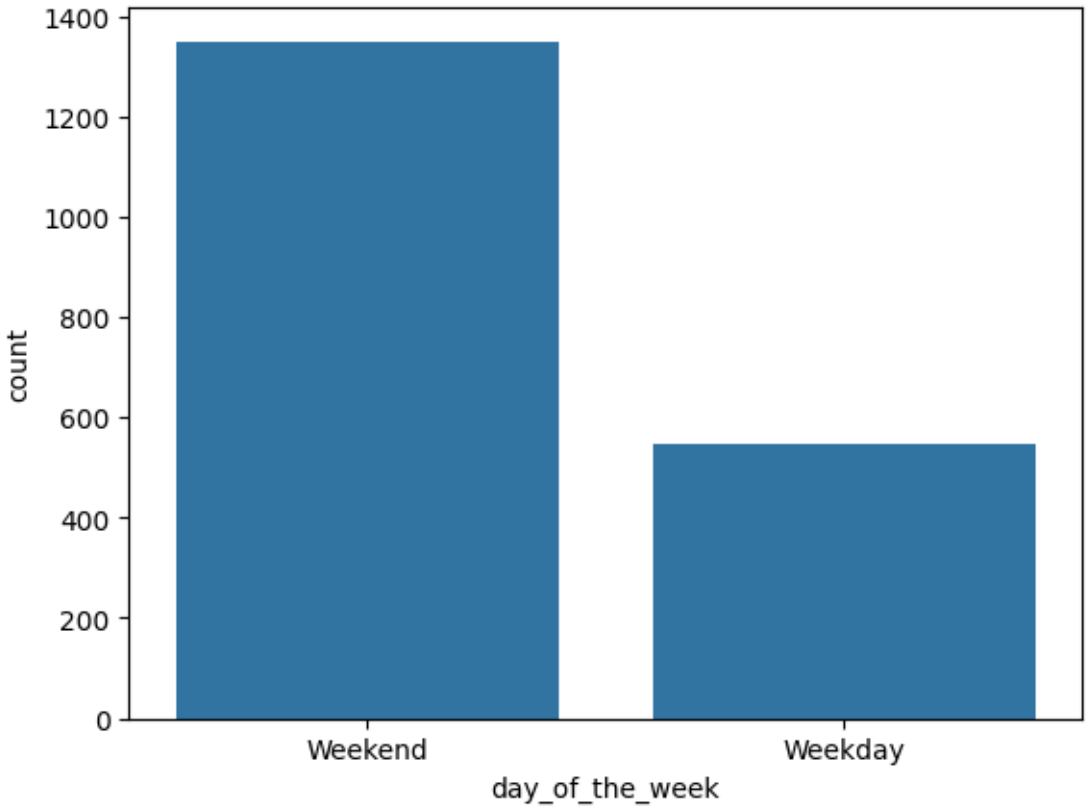
Code Output 6. Unique Values for the Day of the Week column

Weekend, Weekday

Preliminary Observations for Day of the Week

For the given Day of the Week categorical variable, there are two unique values, which are Weekend and Weekday. The dataset does not further differentiate, separate, or categorize the data to show individual days, such as Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, and Sunday.

Graph 4. Day of the Week Bar Graph



Observations on Day of the Week

There is a Weekend dominance in the graph, with orders near 1400. This amount is about double the number of orders during Weekdays. Thus, the dataset is heavily skewed toward Weekend activity. Due to higher demand on Weekends, this could be possibly due to leisure time, social gatherings, or dining out. In contrast, the lower Weekday demand could imply routine work schedules or less spending during Weekdays. The graph suggests that more staffing availability, resource allocation, and marketing efforts should be enhanced for Weekdays.

Rating (See Appendix for Code Snippets)

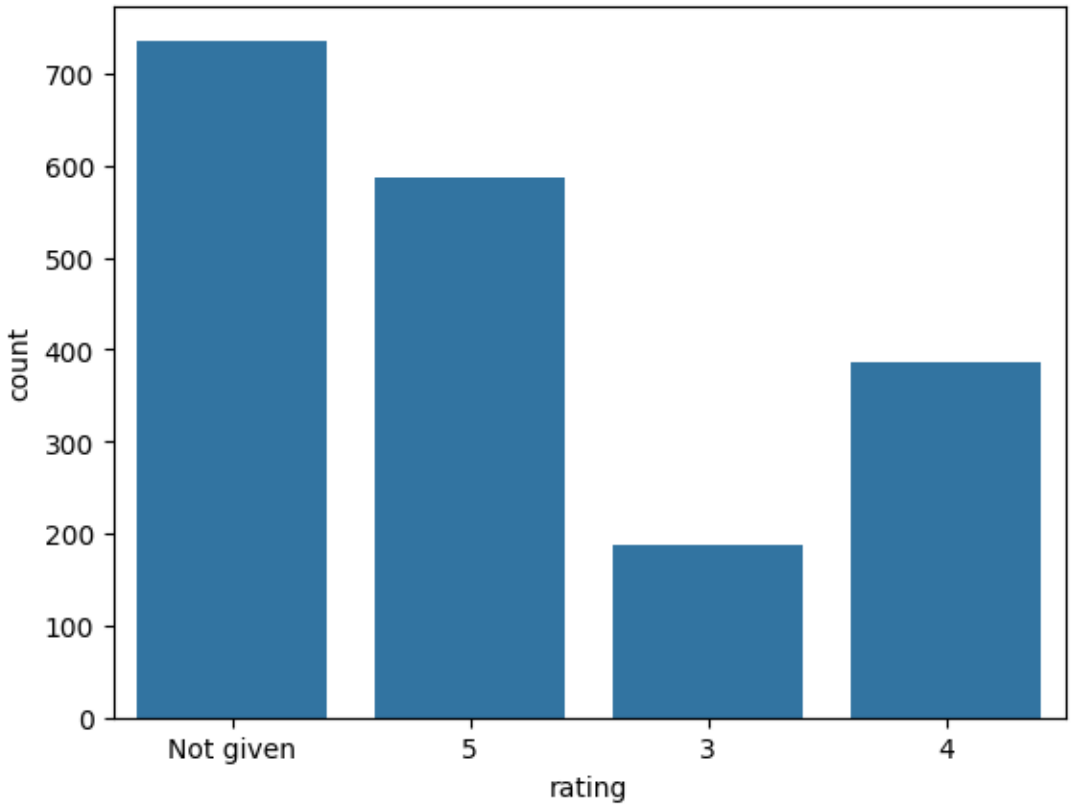
Code Output 7. Unique values for the Rating column

'Not given,' '5,' '3,' '4'

Preliminary Observations for Rating

For the given Rating variable, there are four unique values. They are Ratings of "Not given", "5," "3", and "4." The dataset does not include records for Ratings of "1" or "2," which could indicate that unsatisfied customers chose not to leave a low "1" or "2" Rating, or that most customers were mainly satisfied with the service provided.

Graph 5. Rating Bar Graph



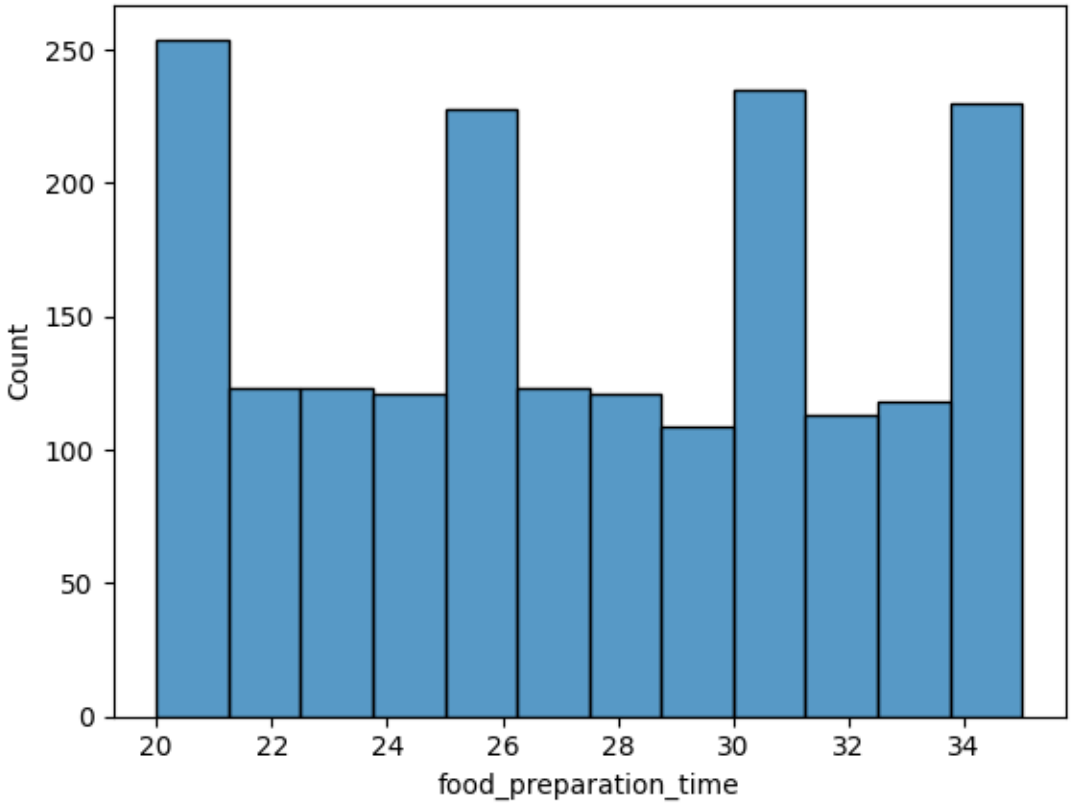
Observations and Implications on Rating Bar Graph

The graph shows a high dominance of "Not given" Ratings, indicating a large proportion of customers or data points without a Rating. As mentioned earlier, this could imply missing data or a significant number of customers abstaining from leaving Ratings. A Rating of "5" is the second most common, with around 600 Ratings, suggesting a strong inclination toward high satisfaction or approval among customers

who chose to leave a Rating. There are about 390 Ratings of "4", indicating moderate satisfaction, but not being thoroughly impressed. The lowest Rating, "3," which is about 190 Ratings, indicates possibly neutral or slightly negative satisfaction from a few customers. There is a positive bias, as Ratings of "5" and "4" account for a substantial share of responses, suggesting favorable feedback. There is a skewed distribution, with fewer "3" and "4" Ratings and a majority of "5" Ratings. Again, this is evidence that there should be incentives for customers to leave Ratings to improve service quality and identify weak points.

Food Preparation Time (See Appendix for Code Snippets)

Graph 6. Food Preparation Time Histogram

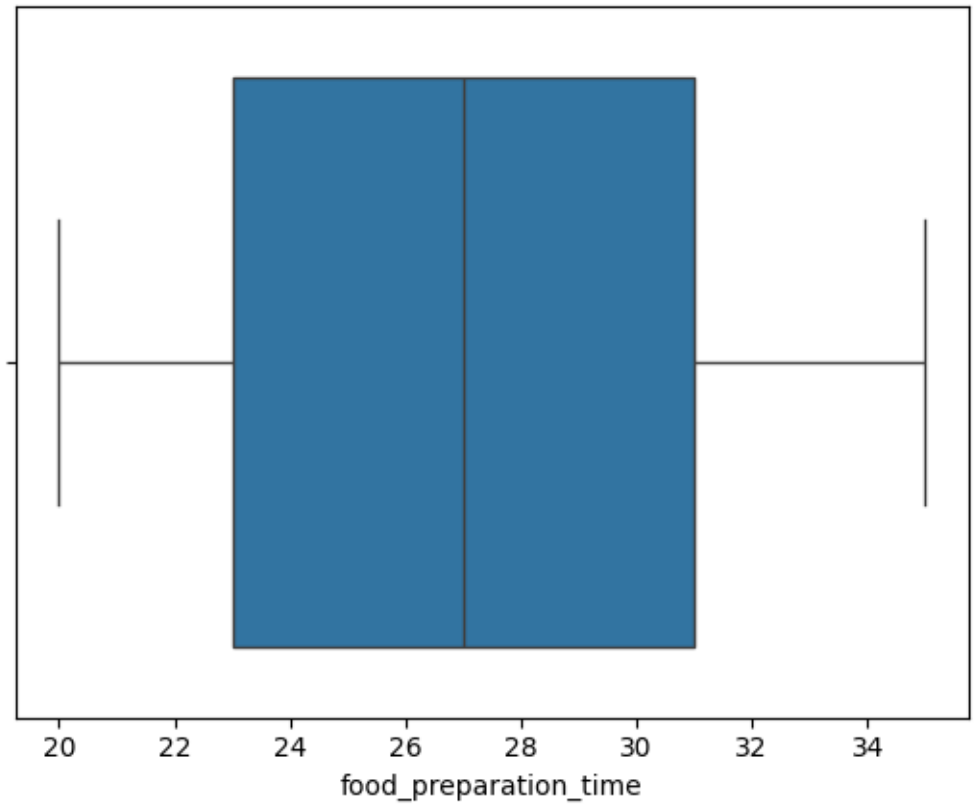


Observations on Food Preparation Time Histogram

The histogram shows that the distribution of Food Preparation Time appears multimodal, with several peaks and a relatively balanced distribution. The histogram shows that these multimodal peaks are noticeable, showing around 20, 26, 30, and 34 minutes for Food Preparation Times. They also suggest different Food Preparation

Times, likely due to different cuisines or specific food products. The histogram shows that the distribution of Food Preparation Time is not perfectly symmetric, suggesting that Food Preparation Times cluster around several common durations. These common durations range from 22 to 33 minutes. These time durations could reflect different meal types or kitchen workflows.

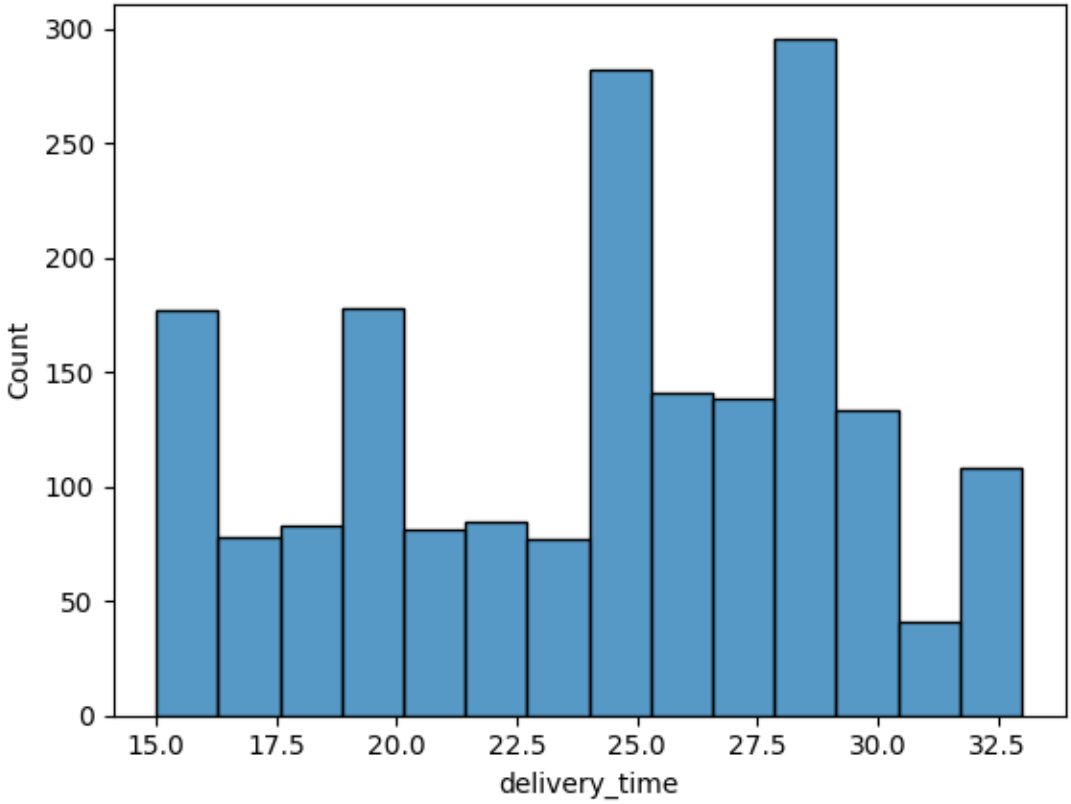
Graph 7. Food Preparation Time Boxplot



The boxplot shows that the data is relatively symmetric, though slightly right-skewed. Food Preparation Time has a lower quartile 1 bound of 23 minutes, and an upper quartile 3 bound of 31 minutes. Given the lower quartile (1) and upper quartile (3) bounds, most Food Preparation Times fall between 23 and 31 minutes. The median for Food Preparation Times is about 27 minutes. The boxplot indicates a minimum of about 20 minutes, and a maximum of about 35 minutes. As previously referenced, there should be kitchen efficiency improvements and menu streamlining to reduce Food Preparation Times.

Delivery Time (See Appendix for Code Snippets)

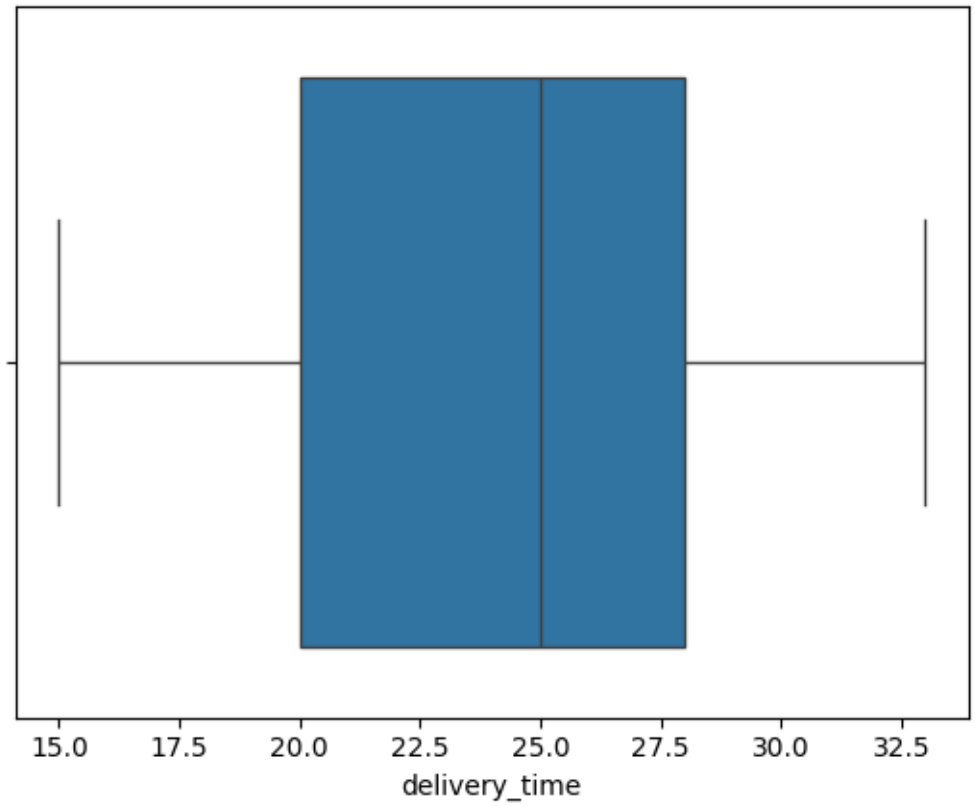
Graph 8. Delivery Time Histogram



Observations on Delivery Time Histogram

The histogram shows that the distribution of Delivery Times appears slightly right-skewed. The histogram shows noticeable multi-structured peaks clustering around Delivery Times of 25 and 28 minutes. The peaks also indicate that standard delivery durations exceed 28 minutes. There are also secondary peaks, indicating Delivery Times of about 15 and 20 minutes. There is also a wide spread range in the histogram, suggesting high variability in service speed. These findings indicate that Delivery Times range from about 28 minutes to about 25 minutes.

Graph 9. Delivery Time Boxplot



Observations on Delivery Time Boxplot

The boxplot shows a central tendency with a median of around 25 minutes in Delivery Times. There is a lower quartile 1 bound of about 20 minutes, and an upper quartile 3 bound of about 28 minutes. Given the lower quartile (1) and upper quartile (3) bounds, most Delivery Times fall between 20 and 28 minutes. The boxplot indicates a minimum of about 15 minutes, and a maximum of about 33 minutes. Likewise, based on the suggestions mentioned above, Delivery Times for orders could decrease significantly, with median times below 25 minutes. Based on the boxplot, there is a need to optimize routes, improve hotspot delivery, and enhance customer communication to identify and reduce delays and improve overall satisfaction.

Univariate Analysis (Answers to requested business application research plans)

Top 5 Restaurants in terms of the Orders received (See Appendix for Code Snippet)

Table 6. Top 5 Restaurants with the highest Number of Orders

	count
restaurant_name	
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

Observations on the Top 5 Restaurants according to the Number of Orders

The top 5 restaurants by the Number of Orders received are Shake Shack (e.g., 219 orders), The Meatball Shop (e.g., 132 orders), Blue Ribbon Sushi (e.g., 119 orders), Blue Ribbon Fried Chicken (96 e.g., orders), and Parm (68 e.g., orders). There is also a brand relationship between Blue Ribbon Sushi and Blue Ribbon Fried Chicken, where one Restaurant serves Japanese cuisine and the other American cuisine.

Most popular Cuisine Type on Weekends (See Appendix for Code Snippet)

Table 7. Most Popular Cuisine Type on Weekends

	count
cuisine_type	
American	415

Observations on the Most Popular Cuisine Type on Weekends

The most popular Cuisine Type on Weekends is American, with 415 orders. This finding indicates a strong customer inclination for American-style restaurants. These American restaurants account for the majority of overall Revenue on Weekends, and most customer demographics prefer American cuisine.

Percentage of Orders above $20 (See Appendix for Code Snippets)

***Code Output. 8* Number and Percentage of Total Orders above $20**

The number of total orders that cost above 20 dollars is: 555
Percentage of orders above 20 dollars: 29.24 %

Observations on the Percentage of Orders above than $20

The output indicates that the total Number of Orders costing more than $20 is 555, and that the percentage of such orders is 29.24%. Conversely, this implies that the majority of orders in the dataset cost $20 or less, accounting for 70.76%. Restaurants with orders over $20 are significant contributors to revenue optimization. There should be available marketing strategies to influence upselling or bundling to encourage customers in this bracket. About 1 in 3 orders are high-value orders of above $20.

Mean Order Delivery Time (See Appendix for Code Snippet)

Code Output. 9 Mean Delivery Time

The mean delivery time for this dataset is 24.16 minutes.

Observations on Mean Order Delivery Time

This finding implies that, on average, customers do not wait more than 30 minutes for their Order to be delivered. In addition, this indicates that the Delivery Time is typical and acceptable.

Top 3 Most Frequent Customers eligible for 20% discount vouchers (See Appendix for Code Snippet)

Table 8. Top 3 Most Frequent Customers

	count
customer_id	
52832	13

	count
customer_id	
47440	10
83287	9

Observations on the Top 3 Most Frequent Customers Table

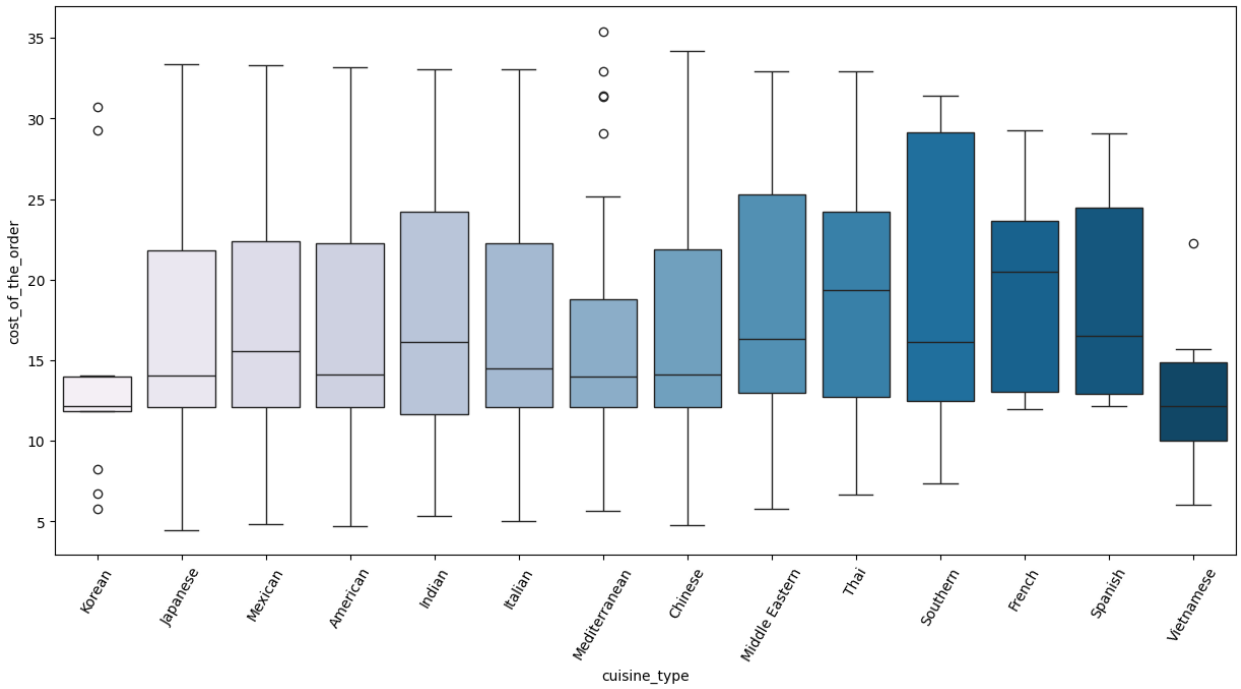
The top 3 most frequent customers eligible for 20% discount vouchers are the customers with IDs 52832, 47440, and 83287, with order amounts of 13, 10, and 9, respectively. This finding implies that these customers are highly engaged, loyal, and satisfied with the FoodHub food aggregator. Therefore, they must continue to contribute to the company's overall Revenue.

Multivariate Analysis

A multivariate analysis was conducted to examine the nature, distribution, and implications of key numerical variables, as well as the relations between numerical and categorical variables in the dataset. Recall that the median is the middle value in a dataset, which splits the data into two equal halves. Also, recall that the terms quartile 1 and quartile 3 indicate that the data are located in the lower 25% and upper 75% of the data shown in the following boxplots, where both lower and upper bounds represent the majority of values for distinct numerical variables.

Cuisine Type vs. Cost of the Order (See Appendix for Code Snippet)

Graph 10. Cuisine Type vs. Cost of the Order Boxplots



Observations and Implications on Cuisine Type vs. Cost of the Order Boxplots

Southern, Japanese, and French cuisines are higher-cost tier cuisines. Southern cuisine has a lower quartile 1 bound of about $11, an upper quartile 3 bound of about $29, and a median of about $15. Given the lower quartile (1) and upper quartile (3) bounds, most Southern cuisine costs between $11 and $29. Southern cuisine has a minimum of about $7, and a maximum of about $31. Japanese cuisine has a lower quartile 1 bound of about $12, an upper quartile 3 bound of about $22, and a median of about $15. Given the lower quartile (1) and upper quartile (3) bounds, most Japanese cuisine costs between $12 and $22. Japanese cuisine has a minimum of about $5, and a maximum of about $34. French cuisine has a lower quartile 1 bound of about $13, an upper quartile 3 bound of about $22, and a median of about $22. Given the lower quartile (1) and upper quartile (3) bounds, most French cuisine costs between $13 and $22. French cuisine has a minimum of about $13, and a maximum of about $30. The higher-order prices in these Cuisine Types may indicate premium options, specialty dishes, or bundled orders.

Italian, Mediterranean, American, Mexican, Indian, and Spanish cuisines are moderate-cost tier cuisines. Italian cuisine has a lower quartile 1 bound of about $13, an upper quartile 3 bound of about $21, and a median of about $15. Given the lower

quartile (1) and upper quartile (3) bounds, most Italian cuisine costs between $13 and $21. Italian cuisine has a minimum of about $6, and a maximum of about $34. Mediterranean cuisine has a lower quartile 1 bound of about $13, an upper quartile 3 bound of about $18, and a median of about $14. Given the lower quartile (1) and upper quartile (3) bounds, most Mediterranean cuisine costs between $13 and $18. Mediterranean cuisine has a minimum of about $6, and a maximum of about $25. American cuisine has a lower quartile 1 bound of about $13, an upper quartile 3 bound of about $22, and a median of about $14. Given the lower quartile (1) and upper quartile (3) bounds, most American cuisine costs between $13 and $22. American cuisine has a minimum of about $5, and a maximum of about $34. Mexican cuisine has a lower quartile 1 bound of about $13, an upper quartile 3 bound of about $23, and a median of about $16. Given the lower quartile (1) and upper quartile (3) bounds, most Mexican cuisine costs between $13 and $23. Mexican cuisine has a minimum of about $6, and a maximum of about $34. Indian cuisine has a lower quartile 1 bound of about $12, an upper quartile 3 bound of about $24, and a median of about $16. Given the lower quartile (1) and upper quartile (3) bounds, most Indian cuisine costs between $12 and $24. Indian cuisine has a minimum of about $6, and a maximum of about $34. Spanish cuisine has a lower quartile 1 bound of about $14, an upper quartile 3 bound of about $24, and a median of about $16. Given the lower quartile (1) and upper quartile (3) bounds, most Spanish cuisine costs between $14 and $24. Spanish cuisine has a minimum of about $13, and a maximum of about $29. American cuisine has medians that are mid-range and moderately spread across the data, suggesting balanced pricing for its products. As for Mexican and Indian cuisine, the data suggests consistent pricing across their products around a balanced median. Mediterranean, Italian, and Spanish cuisines show less price variability, indicating consistency.

Korean, Vietnamese, Middle Eastern, Chinese, and Thai cuisines are lower-cost tier cuisines. Korean cuisine has a lower quartile 1 bound of about $12, an upper quartile 3 bound of about $14, and a median of about $13. Given the lower quartile (1) and upper quartile (3) bounds, most Korean cuisine costs between $12 and $14. Korean cuisine has a minimum of about $13, and a maximum of about $14. Vietnamese cuisine has a lower quartile 1 bound of about $10, an upper quartile 3 bound of about $12, and a median of about $11. Given the lower quartile (1) and upper quartile (3) bounds, most Vietnamese cuisine costs between $10 and $12. Vietnamese cuisine has a minimum of about $10, and a maximum of about $14. Middle

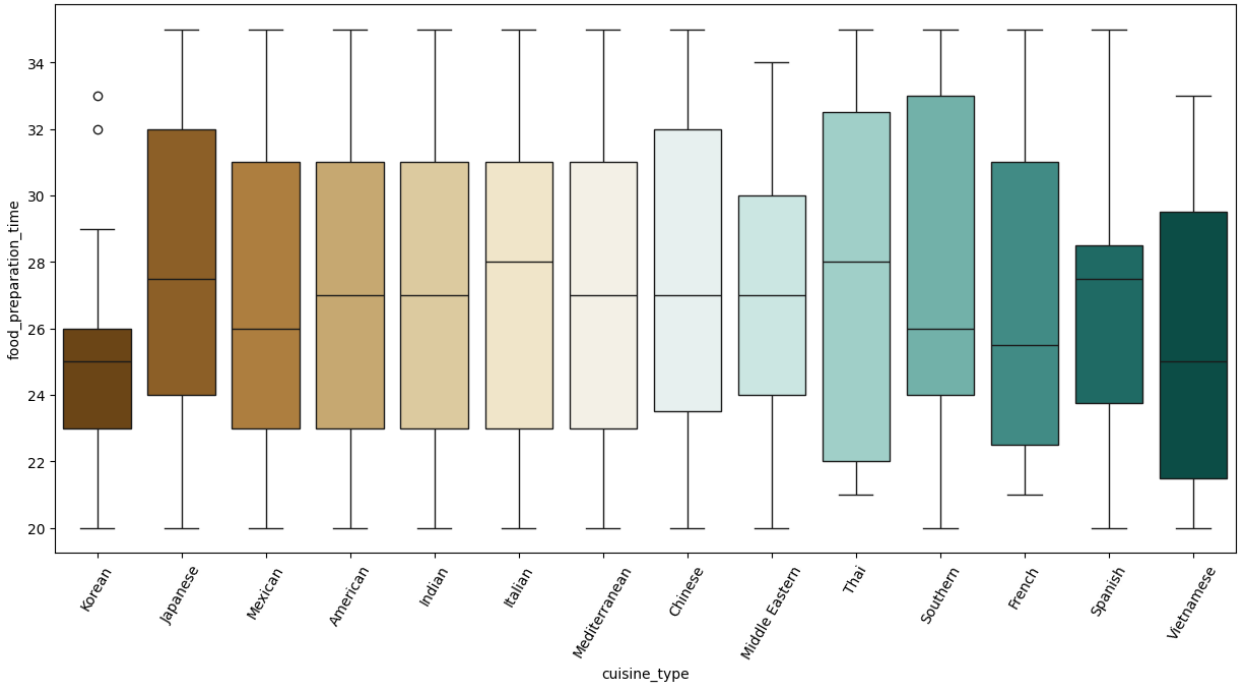
Eastern cuisine has a lower quartile 1 bound of about $12, an upper quartile 3 bound of about $24, and a median of about $17. Given the lower quartile (1) and upper quartile (3) bounds, most Middle Eastern cuisine costs between $12 and $24. Middle Eastern cuisine has a minimum of about $6, and a maximum of about $33. Chinese cuisine has a lower quartile 1 bound of about $14, an upper quartile 3 bound of about $22, and a median of about $15. Given the lower quartile (1) and upper quartile (3) bounds, most Chinese cuisine costs between $14 and $22. Chinese cuisine has minimum of about $5, and a maximum of about $34. Thai cuisine has a lower quartile 1 bound of about $13, an upper quartile 3 of about $24, and a median of about $19. Given the lower quartile (1) and upper quartile (3) bounds, most Thai cuisine costs between $13 and $24. Thai cuisine has a minimum of about $7, and a maximum of about $33. Korean and Vietnamese cuisines have low medians and narrow boxes, indicating lower variability in their order prices. These cuisines suggest consistent, lower variability, and affordable prices for their products.

Furthermore, there are outlier data points visible for the following cuisines. There are three outlier points below the lower bound and past the minimum Cost, and two outlier points above the maximum values for Korean cuisine. There are also several outlier data points for Mediterranean cuisine, all located above the upper bound, past the maximum value. Finally, one data point is considered an outlier for Vietnamese cuisine because it exceeds the upper bound.

The overall implication is that most customers prefer lower-cost cuisines to higher-cost ones. Marketing efforts should focus on both premium and budget Cuisine Types.

Cuisine Type vs. Food Preparation Time (See Appendix for Code Snippet)

Graph 11. Cuisine Type vs. Food Preparation Time Boxplots



Observations and Implications on Cuisine Type vs. Food Preparation Time Boxplots

Korean, Vietnamese, Middle Eastern, and Chinese cuisines are faster prep-tier cuisines. They show lower medians, while also indicating consistently shorter Food Preparation Times. Korean cuisine has a lower quartile 1 bound of about 23 minutes, an upper quartile 3 bound of about 26 minutes, and a median of about 25 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Korean dishes take between 23 and 26 minutes to prepare. Korean cuisine takes a minimum of about 20 minutes, and a maximum of about 29 minutes. Vietnamese cuisine has a lower quartile 1 bound of 22 minutes, an upper quartile 3 bound of about 29 minutes, and a median of about 25 minutes. Given the lower quartile (1) and the higher quartile (3) bounds, most Vietnamese dishes take between 22 and 29 minutes to prepare. Vietnamese cuisine takes a minimum of about 20 minutes and a maximum of about 32 minutes. Middle Eastern cuisine has a lower quartile 1 bound of 24 minutes, an upper quartile 3 bound of about 30 minutes, and a median of about 27 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Middle Eastern dishes take between 24 and 30 minutes to prepare. Middle Eastern cuisine takes a minimum of about 20 minutes and a maximum of about 34 minutes. Chinese cuisine has a lower quartile 1 bound of about 23 minutes, an upper quartile 3 bound of about 32 minutes, and a median of about 27 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Chinese dishes take between 23 and 32 minutes to prepare. Chinese cuisine takes a minimum of about 20 minutes, and a maximum of about 35 minutes.

Mexican, American, Indian, Mediterranean, Spanish, and Italian cuisines are moderate-speed prep-tier cuisines. They have middle-range medians and compact boxes, which are generally predictable prep durations. Mexican cuisine has a lower quartile 1 bound of about 23 minutes, an upper quartile 3 bound of about 31 minutes, and a median of about 26 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Mexican dishes take between 23 and 31 minutes to prepare. Mexican cuisine takes a minimum of about 20 minutes and a maximum of about 35 minutes. American cuisine has a lower quartile 1 bound of about 23 minutes, an upper quartile 3 bound of about 31 minutes, and a median of about 27 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most American dishes take between 23 and 31 minutes to prepare. American cuisine takes a minimum of about 20 minutes, and a maximum of about 35 minutes. Indian cuisine has a lower quartile 1 bound of about 23 minutes, an upper quartile 3 bound of about 31 minutes, and a median of about 27 minutes. Given the lower quartile (1) and the higher quartile (3) bounds, most Indian dishes take between 23 and 31 minutes to prepare. Indian cuisine takes a minimum of about 20 minutes, and a maximum of about 35 minutes. Mediterranean cuisine has a lower quartile 1 bound of about 23 minutes, an upper quartile 3 bound of about 31 minutes, and a median of about 27 minutes. Given the lower quartile (1) and the higher quartile (3) bounds, most Mediterranean dishes take between 23 and 31 minutes to prepare. Mediterranean cuisine takes a minimum of about 20 minutes, and a maximum of about 35 minutes. Spanish cuisine has a lower quartile 1 bound of about 24 minutes, an upper quartile 3 bound of about 29 minutes, and a median of about 28 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Spanish dishes take between 24 and 29 minutes to prepare. Spanish cuisine takes a minimum of about 20 minutes, and a maximum of about 35 minutes. Italian cuisine has a lower quartile 1 bound of about 23 minutes, an upper quartile 3 bound of about 31 minutes, and a median of 28 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Italian dishes take between 23 and 31 minutes to prepare. Italian cuisine takes a minimum of about 20 minutes and a maximum of about 35 minutes.

Japanese, French, Thai, and Southern cuisines are among the longest-prep cuisines. Japanese cuisine has a lower quartile 1 bound of about 24 minutes, an upper quartile 3 bound of about 32 minutes, and a median of about 28 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Japanese dishes take between 24 and 32 minutes to prepare. Japanese cuisine has a minimum of about 20 minutes, and a maximum of about 35 minutes. French cuisine has a lower quartile 1 bound of about 23 minutes, an upper quartile 3 bound of 31 minutes, and a median of

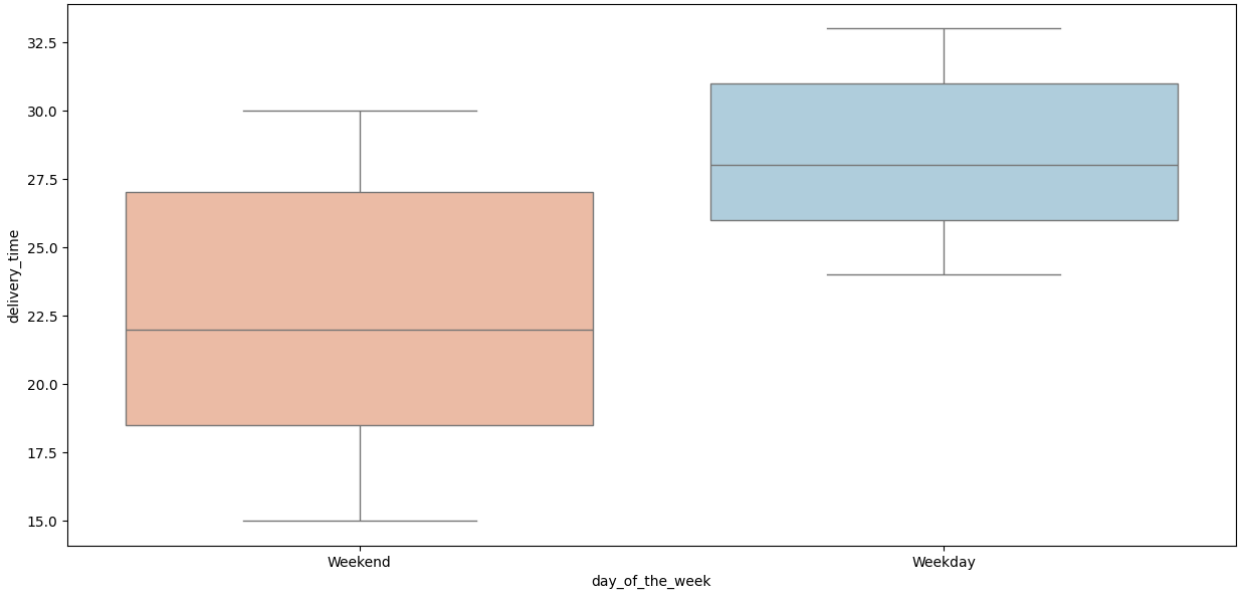
about 24 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most French dishes take between 23 and 31 minutes to prepare. French cuisine takes a minimum of about 21 minutes, and a maximum of about 35 minutes. Thai cuisine has a lower quartile 1 bound of about 22 minutes, an upper quartile 3 bound of about 32 minutes, and a median of about 28 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Thai dishes take between 22 and 32 minutes to prepare. Thai cuisine takes a minimum of about 21 minutes, and a maximum of about 35 minutes. Southern cuisine has a lower quartile 1 bound of 24 minutes, an upper quartile 3 bound of about 32 minutes, and a median of about 26 minutes. Given the lower quartile (1) and the upper quartile (3) bounds, most Southern dishes take between 24 and 32 minutes to prepare. Southern cuisine takes a minimum of about 20 minutes, and a maximum of about 35 minutes.

In contrast to all other cuisines, Thai and French cuisines have longer minimum Food Preparation Times of about 21 minutes, while Korean, Middle Eastern, and Vietnamese cuisines have shorter maximum Food Preparation Times of about 29 minutes, 34 minutes, and 32 minutes, respectively. There are also two data point outliers above the upper bound, beyond the maximum value for Korean cuisine, suggesting that the Food Preparation Time for those data points may be due to premium dish specialties, bundled orders, or fewer staff.

The overall implications of Cuisine Type and Food Preparation Time are to promote faster-prep cuisines through quick-delivery marketing campaigns, while also streamlining long-prep time cuisines to lower variability.

Day of the Week vs. Delivery Time (See Appendix for Code Snippet)

Graph 12. Day of the Week vs. Delivery Time Boxplots



Observations and Implications on Day of the Week vs. Delivery Time Boxplots

Based on the Weekend boxplot, the median Delivery time (e.g., about 22.5 minutes) is substantially lower than on Weekends. The lower quartile 1 bound is about 18 minutes, and the higher quartile 3 bound is about 27.5 minutes. Given the lower quartile (1) and upper quartile (3) bounds, most Weekend Delivery Times fall between 18 minutes and 27.5 minutes. The minimum is about 15 minutes, and the maximum is about 30 minutes. It is possible to infer that Weekend deliveries are quicker on average due to more staffing across restaurants or equivocal demand.

Based on the Weekday boxplot, the median Delivery Time (e.g., about 27.5 minutes) is substantially higher than on Weekends. The lower quartile 1 bound is about 27.5 minutes, and the upper quartile 3 bound is about 30 minutes. Given the lower quartile (1) and upper quartile (3) bounds, most Weekday Delivery Times fall between 27.5 and 30 minutes. The minimum is about 25 minutes, and the maximum is about 32.5 minutes. It is possible to infer that Weekday deliveries take longer than Weekend deliveries, but they are more consistent and may take longer due to traffic caused by workday congestion.

The overall implications are to implement more Weekday food staff employees and promote faster Weekend Delivery Times to attract more customers.

Revenue Generated Across the Top 14 Performing Restaurants (See Appendix for Code Snippet)

Table 9. Top 14 Performing Restaurants by Revenue

	cost_of_the_order
restaurant_name	
Shake Shack	3579.53
The Meatball Shop	2145.21
Blue Ribbon Sushi	1903.95
Blue Ribbon Fried Chicken	1662.29
Parm	1112.76
RedFarm Broadway	965.13
RedFarm Hudson	921.21
TAO	834.50
Han Dynasty	755.29
Blue Ribbon Sushi Bar & Grill	666.62
Rubirosa	660.45
Sushi of Gari 46	640.87
Nobu Next Door	623.67
Five Guys Burgers and Fries	506.47

Observations and Implications on the Top 14 Performing Restaurants Table

Based on the restaurant revenue table, Shake Shack is the top-performing Restaurant in the entire dataset, with $3,579.53 in Revenue, approximately 67% higher than The Meatball Shop (e.g. $2145 in Revenue), the second-most-performing Restaurant. Aside from Shake Shack and The Meatball Shop, Blue Ribbon Sushi (e.g., $1903.95 in Revenue), Blue Ribbon Fried Chicken (e.g., $1662.29 in Revenue), and Parm (e.g., $1112.76 in Revenue) are ranked 3rd, 4th, and 5th, respectively. The restaurants performing between the top and lower performing with their respective Revenue are: RedFarm Broadway (e.g., $965.13 in Revenue), RedFarm Hudson (e.g., $921.21 in Revenue), TAO (e.g., $834.50 in Revenue), Han Dynasty (e.g., $755.29 in Revenue), Blue Ribbon Sushi Bar & Grill (e.g., $666.62 in Revenue), Rubirosa (e.g., $660.45 in Revenue), and Sushi of Gari 46 (e.g., $640.87 in Revenue). Restaurants ranking at the lower end of the table, such as Nobu Next Door (e.g., $623.67 in

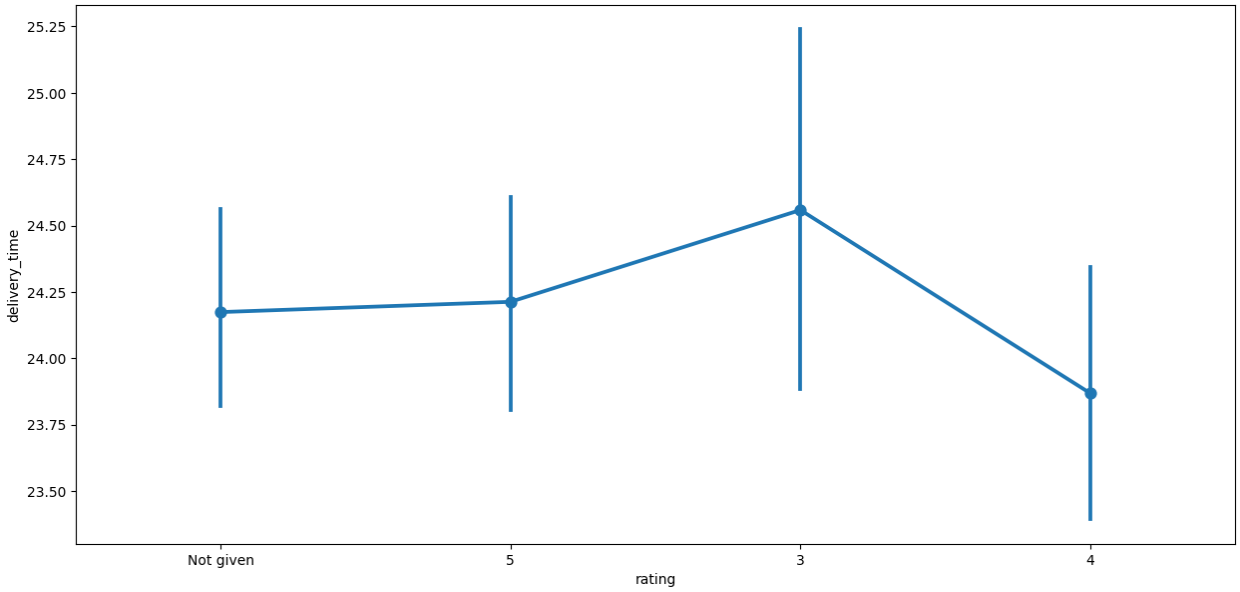
Revenue) and Five Guys Burgers and Fries (e.g., $506.47), have the lowest total Revenue, which could be due to fewer orders or a lower average order value.

Again, it is possible to infer a strong branding cluster, as the names Blue Ribbon and RedFarm appear multiple times. In other words, this suggests that Blue Ribbon offers a variety of cuisines, including American and Japanese. In the same way, RedFram may suggest serving the same type of cuisine in different locations. Likewise, some possible high spenders include American, Italian, and Asian cuisines, suggesting popular or premium menu items. On another note, Shake Shack's top revenue performance might be due to higher customer traffic rather than higher prices. In contrast, sushi dishes available in Asian restaurants may reflect higher-quality, exclusive offerings.

The implications for restaurant revenue results include possible revenue dependency risks (reliance on high performing restaurants), brand architecture strategy (focus on multi-location brands such as Blue Ribbon and RedFarm), growth opportunities (focus on mid-tier restaurants such as Parm, TAO, and RedFarm with promotions or operational enhancements to push them into the higher-performing tier), low-performing establishments (utilize strategic alignment, contract restructuring between partnerships, or asset reallocation for higher-yield brands), and benchmarking potential (analyze Shake Shack's factors for benchmark performance to understand better how these characteristics could be carried out for other restaurants).

Rating vs. Delivery Time (See Appendix for Code Snippet)

Graph 13. Rating vs Delivery Time Pointplot



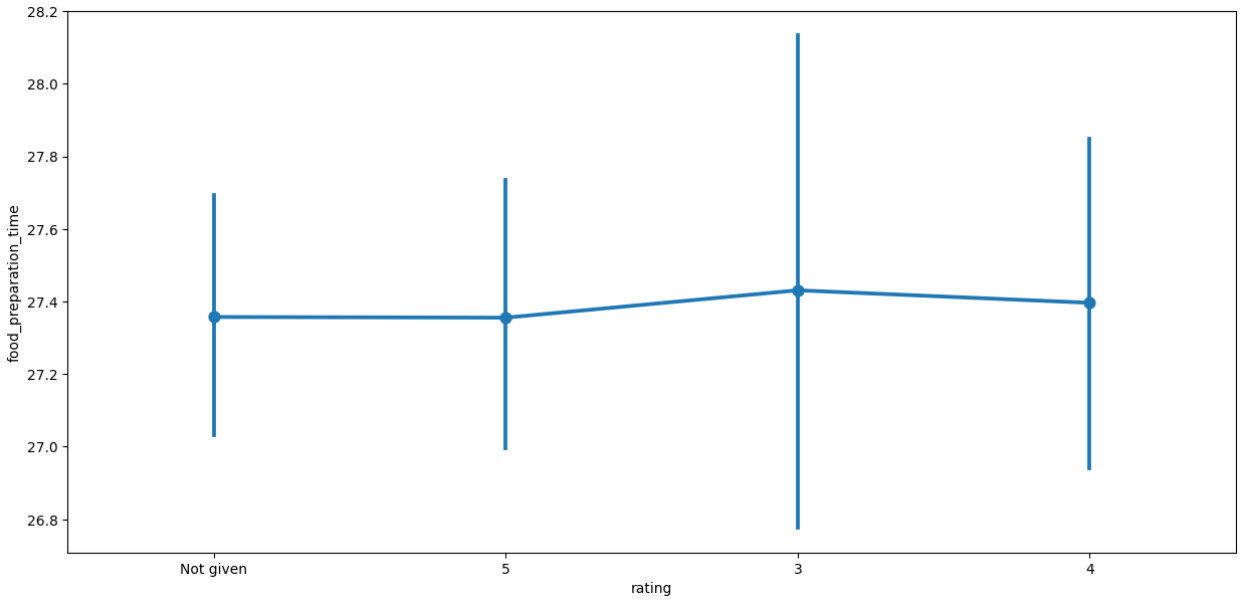
Observations and Implications on Rating vs. Delivery Time Pointplot

The Rating vs. Delivery Time pointplot shows a trend in which Delivery Time increases from a rating of "Not given" (associated average Delivery Time below about 24.25 minutes) to a Rating of 3, then decreases to a Rating of 4. The pointplot further suggests that a Rating of 3 is associated with slower average Delivery Times (about 24.5 minutes), which could indicate poor customer satisfaction or operational delays. A Rating of 4 is associated with the fastest average Delivery Times (e.g., below 24 minutes). A Rating of 5 is associated with moderate average Delivery Times (e.g., slightly below 24.5 minutes). When customers choose not to leave a Rating, the pointplot displays the fastest Delivery Times, which could be due to customers being passive or to the default behavior of not leaving a Rating. The shorter bars for Ratings of “Not Given,” 5, and 4 indicate consistency around the average Delivery Times for each Rating. In contrast, the longer bar for a Rating of 3 indicates greater variability around the average Delivery Times for that Rating.

The implications of the Rating and Delivery Time pointplot suggest operational benchmarking to keep Delivery Time under 25 minutes to retain customer satisfaction, and evidence that Delivery Time is a strong indicator of a customer's Rating.

Rating vs. Food Preparation Time (See Appendix for Code Snippet)

Graph 14. Rating vs. Food Preparation Time Pointplot



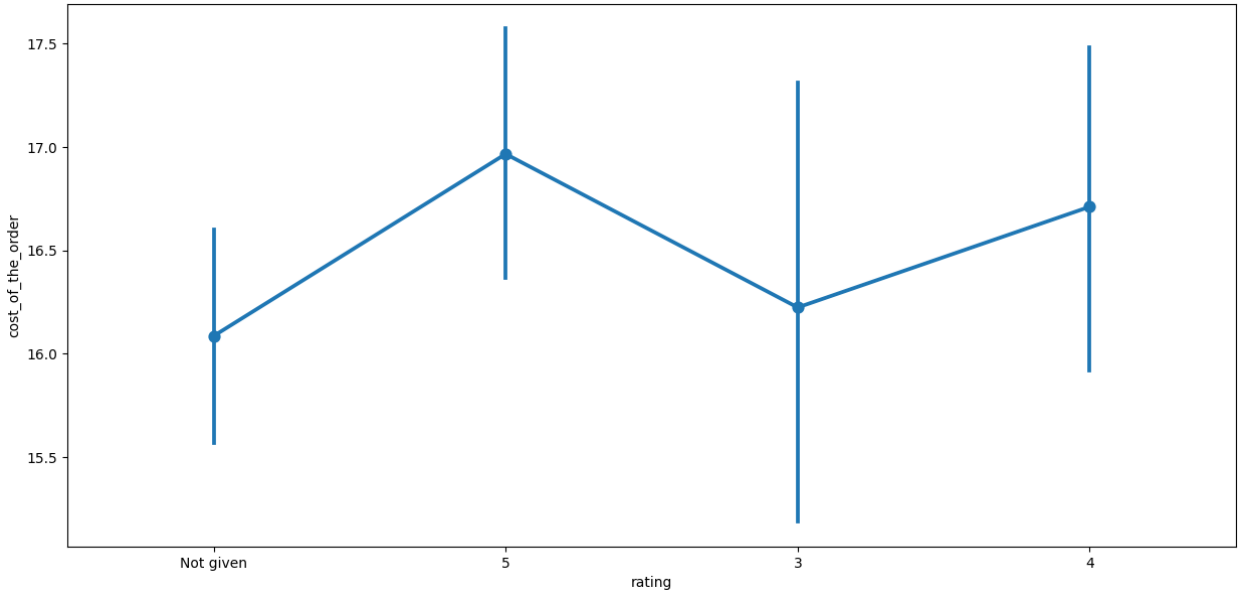
Observations and Implications on Rating vs. Food Preparation Time Pointplot

Similar to the previous Rating vs. Delivery Time pointplot, this Rating vs. Food Preparation Time pointplot shows a trend in which Food Preparation Time increases from a Rating of "Not given" (associated average Food Preparation Time slightly below 27.4 minutes) to a Rating of 3, then decreases to a Rating of 4. Likewise, the pointplot further suggests that a Rating of 3 is associated with slower average Food Preparation Times (about 27.4 minutes), which could imply lower customer satisfaction. A Rating of 4 is associated with similar average Food Preparation Times (e.g., about 27.4 minutes). A Rating of 5 is associated with the fastest average Food Preparation Times (e.g., a little below 27.4 minutes), slightly lower than for a Food Preparation Time associated with a Rating of “Not given.” In the same manner, when customers choose not to leave a Rating, the pointplot displays the fastest Food Preparation Times, which could be due to customers ordering simple orders or being less engaged with the Rating process. The shorter bars for Ratings of “Not Given,” 5, and 4 indicate consistency around the average Food Preparation Times for each Rating. In contrast, the longer bar for a Rating of 3 indicates a larger variability around the average Food Preparation Times for that Rating.

The implications of the Rating and Food Preparation Time pointplot suggest a need for a service quality assessment focused on keeping Food Preparation Time below 27 minutes, as well as a menu staffing review linked to lower Ratings.

Rating vs. Cost of the Order (See Appendix for Code Snippet)

Graph 15. Rating vs. Cost of the Order Pointplot



Observations and Implications on Rating vs. Cost of the Order Pointplot

Similar to the previous Rating vs. Delivery Time and Rating vs. Food Preparation Time pointplots, this Rating vs. Cost of the Order pointplot shows a trend in which Cost of the Order increases from a Rating of "Not given" (associated with the lowest average order costs of about $16) to a Rating 3, then decreases to a Rating 4. A Rating of 3 is associated with average orders of about $16.50, suggesting that medium-priced orders do not always lead to higher customer satisfaction. A Rating of 4 is associated with average order costs of about $16.75, implying that second-to-high-expensive orders indicate more satisfied customers. A Rating of 5 is associated with the highest average order costs, at about $17, suggesting that customers may be more satisfied with higher-priced orders. A "Not given" Rating corresponds to the cheapest costs across orders placed, which could imply lighter, smaller orders. The shorter bars for Ratings of “Not Given,” 5, and 4 indicate consistency around the average Cost of Orders for each Rating. In contrast, the longer bar for a Rating of 3 indicates a larger variability around the average Cost of Orders for that Rating.

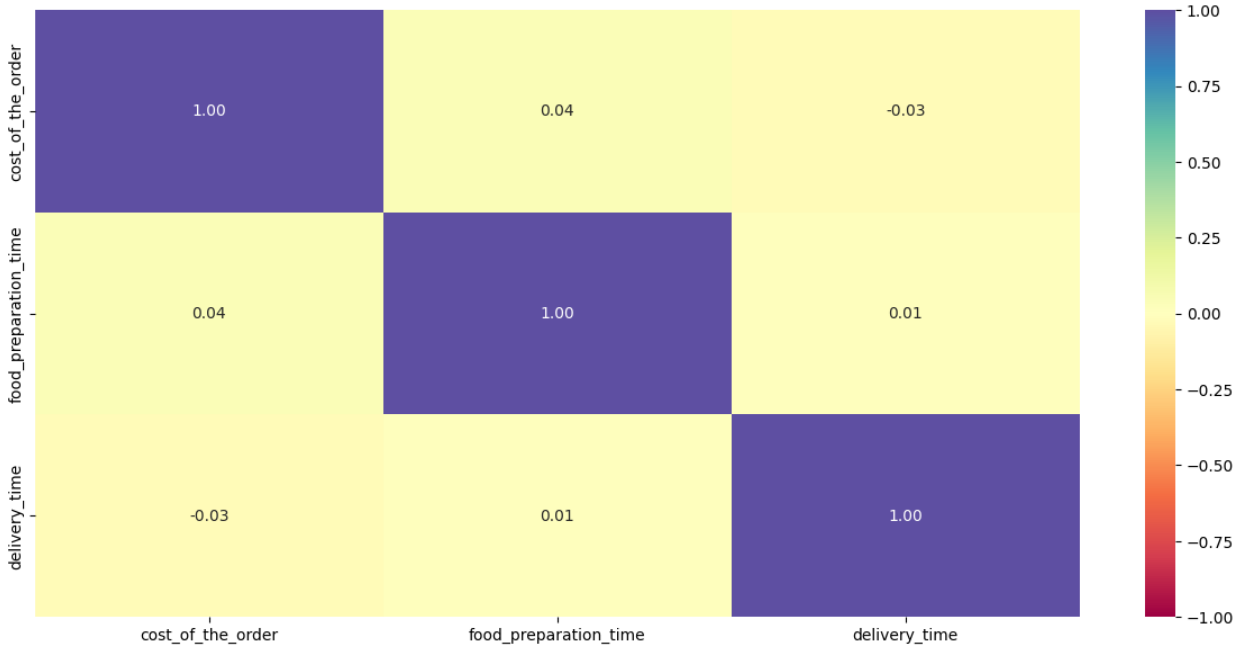
The implications of the Rating and Order Cost pointplot suggest that a pricing strategy could be used to guarantee a premium experience and to implement a continuous improvement cycle that aligns service quality with price tiers. In other words, the data show that restaurants offer a wide variety of high-quality food

products at varying prices and provide options to increase customer satisfaction, such as bundles, seasonal promotions, and limited-time offers.

Correlation among numerical variables (See Appendix for Code Snippet)

Correlation is defined (using a correlation coefficient) as how two numerical variables move together: either in a direct, positive relationship, an inverse, negative relationship, or no correlation. As in, a positive correlation suggests that both variables increase together, a negative correlation indicates that one variable increases while the other decreases, and no correlation indicates that both variables do not increase or decrease together. However, a correlation does not imply causation; it only shows how variables move together.

Graph 16. Heatmap Visualization for Correlations among Numerical Variables



Observations and Implications on the Heatmap Visualization for Correlations among Numerical Variables

The heatmap of the numerical variables in this dataset shows a slight positive correlation (e.g., a correlation coefficient of +0.04) between Cost of the Order and Food Preparation Time. Thus, there is a direct positive correlation: more expensive orders take longer to prepare, but this relationship may be deemed imperceptible. The heatmap also shows a slight negative correlation (e.g., a correlation coefficient of 0.03)

between Cost of the Order and Delivery Time. Thus, there is an inverse relationship: more expensive orders may be delivered more quickly, but this relationship may be deemed inconsequential. The heatmap further shows a slight positive correlation (e.g., a correlation coefficient of +0.01) between Food Preparation Time and Delivery Time. Thus, this could imply a non-meaningful relationship in which longer Food Preparation Times do not reflect slower Delivery Times.

The implications of numerical variables indicate that each (e.g., Cost of the Order, Food Preparation Time, and Delivery Time) is influenced by various operational and customer parameters. As in, a high-cost order does not take longer to prepare or deliver. Moreover, each of these predictive models does not assume strong interdependence among these numerical variables; instead, they are treated as separate features in forecasting models. Again, as previously stated in the solution section, efforts should be made to optimize food preparation and Delivery Times and to develop cost strategies.

Multivariate Analysis (Answers to requested business application research plans)

Restaurants eligible for the Promotional Offer in the advertisement across all Restaurants (See Appendix for Code Snippets)

Table 10. Restaurant Name with Total Rating Count

	restaurant_name	rating
0	Shake Shack	133
1	The Meatball Shop	84
2	Blue Ribbon Sushi	73
3	Blue Ribbon Fried Chicken	64
4	RedFarm Broadway	41

Observation on Restaurant Name with Rating Count Table

These Restaurants not only have the most Ratings but are also the top performers in generating net Revenue for the company.

Table 11. Restaurants eligible for the Promotional Offer

	restaurant_name	average_rating
0	The Meatball Shop	4.51
1	Blue Ribbon Fried Chicken	4.33
2	Shake Shack	4.28
3	Blue Ribbon Sushi	4.22

Observations and Implications on Restaurants eligible for the Promotional Offer table

The Restaurants eligible for the Promotional Offer are The Meatball Shop, Blue Ribbon Fried Chicken, Shake Shack, and Blue Ribbon Sushi. The Meatball Shop has the highest Average Rating of 4.51, making it the standout candidate for a multitude of promotions. As mentioned earlier, the Blue Ribbon Restaurants appear to be a brand, revealing a strong brand relationship across Blue Ribbon-named Restaurants. Specifically, Blue Ribbon Fried Chicken has an Average Rating of 4.33, the second-highest, and Blue Ribbon Sushi has an Average Rating of 4.22, the fourth-highest. These results align with the previous findings of the top five Restaurants with the highest orders, indicating that only the highest-performing Restaurants qualify for this promotional offer. Surprisingly, The Meatball Shop and Blue Ribbon Fried Chicken have higher Ratings than Shake Shack (e.g., an Average Rating of 4.28), even though they were previously ranked second and fourth, respectively, in terms of the most orders. Overall, the Restaurants eligible for this Promotional Offer suggest that they offer high-quality food products and provide customer satisfaction.

The implications for these four Restaurants eligible for the promotional offer are that they should signal high Ratings with sufficient volume, and that marketing efforts should feature them in promotional campaigns to build trust and credibility for FoodHub's smartphone application. FoodHub should leverage cross-promotion across Blue Ribbon restaurant branches to further reinforce cross-promotion and customer recognition. Revenue and demand potential for restaurants, such as Shake Shack and The Meatball Shop, should be leveraged through promotions to capitalize on existing demand and increase average order value, while also positioning these restaurants as core brands across the entire FoodHub marketplace. Strategic positioning efforts should be made to promote these high-performing restaurants and differentiate FoodHub from competing apps, positioning it as a quality-first aggregator with a curated marketplace.

Total Net Revenue generated by the company across all restaurants (See Appendix for Code Snippets)

Table 12. Updated Sample Data Summary Table with Revenue Column added

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time	Revenue
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	20	7.6875
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	23	1.8120
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	28	1.8345
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	15	7.3000
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	24	1.7385

Observations on Updated Sample Data Summary Table

The appropriate follow-up code was generated using the new transformed dataset column table to ensure an accurate updated total net revenue amount, based on the corresponding hierarchical revenue percentages for each Order's Cost.

Code Output. 10 Total Net Revenue

The Total Net Revenue is around 6166.3 dollars.

Observations and Implications on Total Net Revenue Output

The FoodHub food delivery company earned about $6,166 in Revenue, a suitable amount given the amount of data collected and present in the dataset. As the data were collected primarily to show week-time projections between Weekdays and Weekends, this suggests strong performance. High-value orders of $20 or more contribute significantly due to the 25% commission rate. Mid-range orders between $5 and $20 contribute moderately, but at much lower levels. Lower-cost orders of less than $5 do not include a commission rate and do not contribute to overall Revenue. It is preferred to encourage customers to place orders of $20 or more to optimize Revenue, given higher commission rates. Promotions or staff competent in upselling could be used to influence higher-order purchases. Because orders cluster around $10 to $15 (as shown in the previous Cost-of-Order histogram), most Revenue is generated from the 15% commission tier.

The implications of Total Net Revenue across restaurants suggest a tiered commission structure to incentivize customers to place higher-value orders, with efforts to offer upselling and bundling options to reach the $20 threshold. FoodHub should also differentiate pricing strategies to optimize Revenue across tiers. There

should also be plans to incorporate enhanced delivery options, premium visibility, and targeted promotional initiatives to better promote orders with higher cost margins.

The Percentage of Orders that take more than 60 minutes (See Appendix for Code Snippet)

Code Output. 11 Total Number of Orders with Total Delivery Times greater than 60 minutes

The number of total orders that take more than 60 minutes to deliver is: 200
Percentage of orders with delivery times longer than 60 minutes: 10.54 %

Observations and Implications on the code output of the Percentage of orders with Total Delivery Times greater than 60 minutes

According to the output, 200 orders exceeded 60 minutes in Total Delivery Time (e.g., Food Preparation Time plus Delivery Time), representing about 10% of all orders in the dataset. There is an overall efficiency, with nearly 90% of all orders delivered in less than an hour, or more clearly, less than 30 minutes. The FoodHub food delivery company is thus meeting standard customer expectations for the majority of orders. The 10.54% hindrance could be due to delays occurring, possibly due to complicated food preparation leading to longer Food Preparation Times, overwhelmingly higher demand time periods occurring during Weekdays and Weekends, and complications caused by traffic, longer distance deliveries, and whether delivery employees are available to pick up and deliver orders at a given time and location to fulfill orders. Similar to longer Food Preparation Time and Delivery Times, orders with longer Total Delivery Times of more than 60 minutes may lead to lower Ratings and unhappy customers.

The following implications can be made: longer Total Delivery Times can minimize customer trust due to non-compliant delivery schedules. Ratings could plummet, and efforts should be made to lower Delivery Times. FoodHub should investigate reasons for slower Total Delivery Times by evaluating which cuisines take longer to prepare, why Delivery Times are slower on Weekdays, and which Restaurants underperform. FoodHub should also consider using logistics strategies to consolidate high-demand locations rather than dispersed ones, flexible workforce

management approaches, and grouping food preparation tasks with similar cooking or assembly times across Restaurants.

Mean Delivery Times on Weekdays vs. Weekends (See Appendix for Code Snippet)

Code Output. 11 Mean Delivery Times on Weekdays and Weekends

The mean delivery time on weekdays is around 28 minutes
The mean delivery time on weekends is around 22 minutes

Observations and Implications on Mean Delivery Times on Weekdays and Weekends Code Output

The output shows that mean Delivery Times are faster on Weekends than on Weekdays, with a mean difference of 6 minutes. In particular, Weekend Delivery Times are 21% faster. Some inferences that could be made are that there is less traffic on the Weekends, allowing delivery employees to pick up and deliver orders to customers quickly. Even though there are fewer orders on Weekdays, as noted in a previous bar graph, there may be higher demand during office lunch periods or after work commutes when people have dinner. Another inference is that Restaurants might have more employees on Weekends than on Weekdays. Additionally, customers are more likely to place smaller, less costly orders on Weekends, which could lead to shorter Food preparation and Delivery Times. Customer experiences with shorter Delivery Times on Weekends could improve overall satisfaction and Ratings. In contrast, longer Delivery Times on Weekdays could lead to lower Ratings of 3, as shown in the earlier pointplot of Ratings vs. Delivery Times.

The implications for the mean Weekday and Weekend Delivery Times are that Weekday demand is incorporated during peak hours and that methods should be developed to optimize Weekday routes to minimize delays. There should be consistency in Weekday and Weekend Delivery Times to meet customer satisfaction expectations. Again, Restaurants should use optimal kitchen efficiency measures to improve shorter Food Preparation Times on Weekdays.

Conclusions and Recommendations

Taken together, the following key post conclusions and recommendations can be drawn from the FoodHub dataset.

The top-most frequent customers drive disproportionately more Revenue. FoodHub should consider identifying and rewarding the top 20 customers, rather than just three. Weekend demand is stronger than Weekday demand. Promotions and upselling should be featured on Weekends rather than on Weekdays. A Ratings gap where the majority are "Not given" is present in the dataset. There should be efforts to incentivize customers to leave Ratings to improve future data analysis and build customer trust. The FoodHub company's total Revenue comes from three main outlets: Shake Shack, The Meatball Shop, and Blue Ribbon. Cross-brand loyalty from restaurants with familiar brand names (e.g., Blue Ribbon Sushi and Blue Ribbon Fried Chicken, and RedFarm Broadway and RedFarm Hudson) has strong brand recognition across cuisines.

The FoodHub company should focus on featuring restaurants with high Ratings and high Rating counts in marketing campaigns. Delivery Times are longer on Weekdays than on Weekends. Cuisines such as Japanese, French, Thai, and Southern have longer Food Preparation Times. American cuisine is the most preferred cuisine on Weekends. FoodHub should aim to align Weekday deliveries closer to Weekend Delivery Times and standardize longer-prep-time Cuisine Types to enhance customer satisfaction.

There are tiered net revenue commissions: 25% for orders over $20 and 15% for orders between $5 and $20. The majority of order costs fall below $20. FoodHub should consider upselling and bundling to increase average order value. Moreover, FoodHub should incorporate loyalty rewards and exclusive deals to ensure that order values exceed $20.

Order Costs, Food Preparation Time, and Delivery Times have weak correlations. The data science team should continue analyzing each numerical variable using independent optimization measures. The data science team should use predictive techniques based on day type, Cuisine Type, and Delivery Time to better forecast demand and satisfaction. Also, the data science team should further investigate how customers and restaurants, when grouped by Rating behavior, Cuisine Type, and Cost of each Order, can better reveal tailored outreach opportunities based on customer demographics and preferences.

FoodHub should make the following critical strategic positioning efforts. FoodHub should highlight the top-rated Restaurants in the marketplace through promotions to build credibility and differentiate itself from competitors. FoodHub

should reinforce customer retention and acquisition across platforms to enhance networking across each Restaurant, regardless of performance. FoodHub should incorporate business continuity capabilities by scaling logistics and tech infrastructure, while also managing Restaurant diversity and order volume. In simpler terms, FoodHub should hire more delivery drivers, food preparation staff across Restaurants, and marketing staff to promote diverse bundling and promotions to increase average order value. Likewise, as previously mentioned, there should be efforts to modify the FoodHub smartphone app to display route optimization for delivery drivers to reduce Delivery Times, while also using push notifications with follow-up surveys to remind customers to leave a Rating and collect quantitative customer feedback.

In summary, the data-driven findings indicate that the dataset shows FoodHub's growth potential when using the aforementioned customer-centric strategies and operational excellence measures. By following these strategies and measures, FoodHub can strengthen customer loyalty, build restaurant partnerships, and drive viable revenue growth.

Appendix (List of Code Snippets)

Code Snippet 1. Output the first 5 rows found in the Dataset

```
# Return the first 5 rows
df.head()
```

Code Snippet 2. Compute the Shape of the Dataset

```
# Check the shape of the dataset
df.shape
```

Code Snippet 3. Compute the Datatypes of the different column variables

```
#Check the data types of the different columns in the dataset
df.info()
```

Code Snippet 4. Compute the number of Missing Values for each column

```
# Checking for missing values for each column in the dataset
df.isnull().sum()
```

Code Snippet 5. Obtain the summary Descriptive Statistics for the numerical variables

```
# Get the summary statistics of the numerical data
df.describe().round(2).T
```

Code Snippet 6. Count the unique number of times across each value in the Rating column

```
#Count the unique number of times across each value in the rating column
df['rating'].value_counts()
```

Code Snippet 7. Check the Number of unique Order IDs

```
# check number of unique order IDs
df['order_id'].nunique()
```

Code Snippet 8. Check the Number of unique Customer IDs

```
# check number of unique customer IDs
df['customer_id'].nunique()
```

Code Snippet 9. Check the Number of unique Restaurant Names

```
# check number of unique restaurant names
df['restaurant_name'].nunique()
```

Code Snippet 10. Check the Number of unique Cuisine Types

```
# Check unique cuisine type
df['cuisine_type'].nunique()
```

Code Snippet 11. Plot Bar Graph for Cuisine Type

```
# Create a bar graph for cuisine type
plt.figure(figsize = (15,5))
sns.countplot(data = df, x = 'cuisine_type')
```

Code Snippet 12. Plot Histogram for Cost of the Order Variable

```
#Plot Histogram for the cost of order
sns.histplot(data=df,x='cost_of_the_order') ## Histogram for the cost of order
plt.show()
```

Code Snippet 13. Plot Cost of the Order Boxplot

```
#Create a boxplot for cost of order
sns.boxplot(data=df,x='cost_of_the_order') ## Boxplot for the cost of order
plt.show()
```

Code Snippet 14. Check the unique values for the Day of the Week column

```
# Check the unique values for day of the week
df['day_of_the_week'].unique()
```

Code Snippet 15. Plot Day of the Week Bar Graph

```
#Plot bar graph for day of the week
sns.countplot(data = df, x = 'day_of_the_week')
```

Code Snippet 16. Check the unique values for Rating column

```
# Check the unique values for rating
df['rating'].unique()
```

Code Snippet 17. Plot Rating Bar Graph

```
#Plot bar graph for rating
sns.countplot(data = df, x = 'rating')
```

Code Snippet 18. Plot Food Preparation Time Histogram

```
#Plot histogram for food preparation time
sns.histplot(data=df,x='food_preparation_time')
plt.show()
```

Code Snippet 19. Plot Food Preparation Time Boxplot

```
#Plot boxplot for food preparation time
sns.boxplot(data=df,x='food_preparation_time')
plt.show()
```

Code Snippet 20. Plot Delivery Time Histogram

```
#Plot histogram for delivery time
sns.histplot(data=df,x='delivery_time')
plt.show()
```

Code Snippet 21. Plot Delivery Time Boxplot

```
#Plot boxplot for delivery time
sns.boxplot(data=df,x='delivery_time')
plt.show()
```

Code Snippet 22. Obtain the Top 5 Restaurants with the Highest Number of Orders

```
# Get top 5 restaurants with highest number of orders
df['restaurant_name'].value_counts().head(5)
```

Code Snippet 23. Obtain the most popular Cuisine on Weekends

```
# Get most popular cuisine on weekends
df_weekend = df[df['day_of_the_week'] == 'Weekend']
df_weekend['cuisine_type'].value_counts().head(1)
```

Code Snippet 24. Obtain the Percentage of Orders greater than $20

```
# Get orders that cost above 20 dollars
df_greater_than_20 = df[df['cost_of_the_order']>20]

# Calculate the number of total orders where the cost is above 20 dollars
print('The number of total orders that cost above 20 dollars is:', df_greater_than_20.shape[0])

# Calculate percentage of such orders in the dataset
percentage = (df_greater_than_20.shape[0] / df.shape[0]) * 100

print("Percentage of orders above 20 dollars:", round(percentage, 2), '%')
```

Code Snippet 25. Obtain the Mean Order Delivery Time

```
# Get the mean delivery time
mean_del_time = df['delivery_time'].mean()

print('The mean delivery time for this dataset is', round(mean_del_time, 2), 'minutes')
```

Code Snippet 26. Obtain the Top 3 Frequent Customers

```
# Get the counts of each customer_id for top 3 frequent customers
df['customer_id'].value_counts().head(3)
```

Code Snippet 27. Plot Cuisine Type vs Cost of the Order Boxplot

```
# Plot boxplot to show relationship between food preparation time and cuisine type
plt.figure(figsize=(15,7))
sns.boxplot(x='cuisine_type', y='food_preparation_time', data=df, palette='BrBG')
plt.xticks(rotation = 60)
plt.show()
```

Code Snippet 28. Plot Food Preparation Time vs. Cuisine Type Boxplot

```
# Plot boxplot to show relationship between food preparation time and cuisine type
plt.figure(figsize=(15,7))
sns.boxplot(x='cuisine_type', y='food_preparation_time', data=df, palette='BrBG')
plt.xticks(rotation = 60)
plt.show()
```

Code Snippet 29. Plot Day of the Week vs. Delivery Time Boxplot

```
#Plot boxplot to show relationship between day of the week and delivery time
plt.figure(figsize=(15,7))
sns.boxplot(x='day_of_the_week', y='delivery_time', data=df, palette='RdBu')
plt.show()
```

Code Snippet 30. Generate table to show the Revenue generated by the Top 14 Restaurants

```
#Generate table to show revenue generated by top 14 restaurants
df.groupby(['restaurant_name'])['cost_of_the_order'].sum().sort_values(ascending = False).head(14)
```

Code Snippet 31. Plot Rating vs. Delivery Time Pointplot

```
#Plot pointplot to show relationship between rating and delivery time
plt.figure(figsize=(15, 7))
sns.pointplot(x = 'rating', y = 'delivery_time', data = df)
plt.show()
```

Code Snippet 32. Plot Rating vs. Food Preparation Time Pointplot

```
#Plot pointplot to show relationship between rating and food preparation time
plt.figure(figsize=(15, 7))
sns.pointplot(x='rating', y='food_preparation_time', data=df)
plt.show()
```

Code Snippet 33. Plot Rating vs. Cost of the Order Pointplot

```
# Plot pointplot to show relationship between rating and cost of the order
plt.figure(figsize=(15, 7))
sns.pointplot(x='rating', y='cost_of_the_order', data=df)
plt.show()
```

Code Snippet 34. Plot Heatmap for all numerical variables

```
# Plot heatmap to show correlations between cost of order, food preparation time, and delivery time
col_list = ['cost_of_the_order', 'food_preparation_time', 'delivery_time']
plt.figure(figsize=(15, 7))
sns.heatmap(df[col_list].corr(), annot=True, vmin=-1, vmax=1, fmt=".2f", cmap="Spectral")
plt.show()
```

Code Snippet 35. Obtain the Top Restaurants with the most Rating counts

```
# Filter the rated restaurants
df_rated = df[df['rating'] != 'Not given'].copy()

# Convert rating column from object to integer
df_rated['rating'] = df_rated['rating'].astype('int')

# Create a dataframe that contains the restaurant names with their rating counts
df_rating_count = df_rated.groupby(['restaurant_name'])['rating'].count().sort_values(ascending = False).reset_index()
df_rating_count.head()
```

Code Snippet 36. Filter Top Restaurants with Rating Counts more than 50, calculate these Restaurants' Average Rating, and check if the Average Rating is greater than 4

```
# Get the restaurant names that have rating count more than 50
rest_names = df_rating_count[df_rating_count['rating'] > 50]['restaurant_name'] # Complete the code to get the restaurant names having rating count more than 50

# Filter to get the data of restaurants that have rating count more than 50
df_mean_4 = df_rated[df_rated['restaurant_name'].isin(rest_names)].copy()

# Group and sort the restaurant names with their ratings and find the mean rating of each restaurant with descending mean ratings
df_mean_rating = df_mean_4.groupby(['restaurant_name'])['rating'].mean().sort_values(ascending = False).reset_index().dropna()

# Filter restaurants with average rating greater than 4
promotional_restaurants = df_mean_rating[df_mean_rating['rating'] > 4]

#print('The restaurants that are eligible for the promotional offer are:', promotional_restaurants)
print('restaurant_name', df_mean_rating, promotional_restaurants)
```

Code Snippet 37. Create function to determine and compute Net Revenue per Restaurant, and introduce "Revenue" column

```
# Function to determine the revenue
def compute_rev(x):
    if x > 20:
        return x*0.25
    elif x > 5:
        return x*0.15
    else:
        return x*0

# Create a new column called revenue by calculating the revenue across restaurants using the cost each order
df['Revenue'] = df['cost_of_the_order'].apply(compute_rev) # Write the apprpriate column name to compute the revenue
df.head()
```

Code Snippet 38. Obtain Total Net Revenue across Restaurants

```
# Get the total net revenue and print it
total_rev = df['Revenue'].sum() # Write the appropriate function to get the total revenue
print('The net revenue is around', round(total_rev, 2), 'dollars')
```

Code Snippet 39. Calculate and obtain the Percentage of Orders with Delivery Times greater than 60 minutes

```
# Calculate total delivery time and add a new column to the dataframe df to store the total delivery time
df['total_time'] = df['food_preparation_time'] + df['delivery_time']

# Write the code below to find the percentage of orders that have more than 60 minutes of total delivery time
# Get orders that take longer than 60 minutes to deliver
df_total_time_greater_than_60 = df[df['total_time'] > 60]

#Calculate the number of total orders that take longer than 60 minutes to deliver
print('The number of total orders that take more than 60 minutes to deliver is:', df_total_time_greater_than_60.shape[0])

#Calculate percentage of such orders in the dataset
percentage_V2 = (df_total_time_greater_than_60.shape[0] / df.shape[0]) * 100

print('Percentage of orders with delivery times longer than 60 minutes:', round(percentage_V2, 2), '%')
```

Code Snippet 40. Calculate and obtain the Mean Delivery Times for Weekdays and Weekends

```
# Get the mean delivery time on weekdays
print('The mean delivery time on weekdays is around',
      round(df[df['day_of_the_week'] == 'Weekday']['delivery_time'].mean()),
     'minutes')

# Get the mean delivery time on weekends
print('The mean delivery time on weekends is around',
      round(df[df['day_of_the_week'] == 'Weekend']['delivery_time'].mean()),
      'minutes')
```